Applied Water Science (2024) 14:244
https://doi.org/10.1007/s13201-024-02301-4

ORIGINAL ARTICLE

Enhancing spatial prediction of groundwater-prone areas through optimization of a boosting algorithm with bio-inspired metaheuristic algorithms

Seyed Vahid Razavi-Termeh[1] · Abolghasem Sadeghi-Niaraki[1] · Sani I. Abba[2] · Farman Ali[3] · Soo-Mi Choi[1]

Received: 1 August 2024 / Accepted: 4 October 2024 / Published online: 30 October 2024
© The Author(s) 2024

Abstract
Groundwater resources are essential for ensuring a consistent water supply in many regions. Groundwater potential maps (GPMs) can be utilized in many ways to estimate the quantity, quality, and distribution of subsurface water, supporting the decision-making processes of numerous stakeholders. This study contributes to improving the accuracy of GPMs, focusing on implementing Geospatial Artificial Intelligence (GeoAI) models. For this purpose, the accuracy performance of the Extreme Gradient Boosting (XGBoost) algorithm is improved in this study. To do this, two such popular metaheuristic algorithms, i.e., invasive weed optimization (IWO) and biogeography-based optimization (BBO), are integrated into the XGBoost algorithm for modeling and spatial prediction of the areas prone to groundwater. Three models—XGBoost, XGBoost-IWO, and XGBoost-BBO—are implemented within the Python programming environments to execute spatial modeling and generate predictive maps. The evaluation of results unfolds in two stages: model validation and GPM validation. For the training data, the root mean square error (RMSE) and mean absolute error (MAE) indices were 0.165 and 0.121 for XGBoost, 0.13 and 0.087 for XGBoost-IWO, and 0.114 and 0.082 for XGBoost-BBO, respectively. The test data showed similar trends, with XGBoost yielding RMSE and MAE values of 0.424 and 0.295, XGBoost-IWO at 0.416 and 0.287, and XGBoost-BBO at 0.39 and 0.28. XGBoost-BBO, XGBoost-IWO, and XGBoost had a prediction accuracy higher than other models. The respective area under the curve (AUC) of GMPs using receiver operating characteristic (ROC) curves for XGBoost, XGBoost-IWO, and XGBoost-BBO were 81.8 %, 83.1 %, and 83.7 %. Using bio-inspired metaheuristic algorithms, the GPM accuracy rate has improved further. The study of groundwater resources demonstrated how geological feature extraction by GeoAI may help employ advanced techniques.

Keywords Groundwater resource · Spatial prediction · Geospatial artificial intelligence (GeoAI) · Bio-inspired algorithms · Water scarcity

Seyed Vahid Razavi-Termeh and Abolghasem Sadeghi-Niaraki authors contributed equally to this work.

✉ Abolghasem Sadeghi-Niaraki
a.sadeghi@sejong.ac.kr

Seyed Vahid Razavi-Termeh
razavi@sejong.ac.kr

Sani I. Abba
sabba@pmu.edu.sa

Farman Ali
farmankanju@yahoo.com

Soo-Mi Choi
smchoi@sejong.ac.kr

[1] Department of Computer Science & Engineering and Convergence Engineering for Intelligent Drone, XR Research Center, Sejong University, Seoul, Republic of Korea

[2] Department of Civil Engineering, Prince Mohammad Bin Fahd University, Al Khobar, Saudi Arabia

[3] Department of Computer Science and Engineering, School of Convergence, College of Computing and Informatics, Sungkyunkwan University, Seoul 03063, Republic of Korea

Introduction

Groundwater refers to the water that occupies the saturated zone, which is the area where rock fissures or crevices between mineral grains are filled with water (Rahmati et al. 2016). Groundwater, a vital component of the world's freshwater resources and a fundamental natural source for sustaining life (Barzegar et al. 2021; Miraki et al. 2019), serves as a crucial water resource for agricultural, urban, and industrial activities worldwide (Nguyen et al. 2020). Over the last fifty years, there has been a notable expansion in reliance on groundwater (Kayastha et al. 2022), which can significantly contribute to a region's economic development and biodiversity (Nobre et al. 2007). Groundwater is the primary water resource for agriculture, industry, and residential purposes, especially in arid regions with minimal annual precipitation. Agricultural practices account for 42 % of global groundwater utilization (Çelik et al. 2024). According to Azma et al. (2021), using groundwater becomes even more critical in Iran due to the country's primarily arid and semi-arid environment, low surface water supplies, and insufficient wastewater treatment facilities. More than 70% of Iran's population uses groundwater for most of their drinking and domestic water needs (Miraki et al. 2019; Sobhani et al. 2024). Thus, given the fundamental role of water in human life, the direct impact of changes in water resources on society, and the utilization of groundwater in arid and semi-arid regions like Iran as a source of fresh water, there is a need to explore and extract groundwater resources (Zamani et al. 2022).

Geological structures and lithology-related factors like climate, landforms, drainage systems, land use/cover, geological structure, lithology, etc., also play a significant role in determining groundwater potential (Pham et al. 2019; Zhu et al. 2023; Saqr et al. 2024). Besides, the relationships between the factors above appear vital to identifying the groundwater point and quantifying the groundwater resource of a particular area. Because of all the factors that affect groundwater, water resource managers face the complex and time-consuming task of evaluating and locating possible groundwater regions (Miraki et al. 2019; Aderemi et al. 2021; Abd-Elmaboud et al. 2024).

Changes in the amount and quality of groundwater are caused by a combination of factors, including population increase, urbanization, agricultural and industrial operations, wastewater discharge pollution, and chemical fertilizers (Rajaeian et al. 2024; Çelik et al. 2024). Consequently, effective groundwater management is essential to balance various water needs, and its implementation is challenging due to rapid population growth and recurring, prolonged drought periods (Gorelick and Zheng 2015).

To help with efficient water management, studies must be undertaken to determine the possibilities and resources of groundwater (Machiwal et al. 2011; Patra et al. 2018; Ifediegwu 2021; El-Mezayen and El-Hamid 2023). An effective management tool, groundwater potential maps (GPMs), helps managers develop better water management solutions (Nguyen et al. 2020). For consumers and managers, GPMs provide crucial information about groundwater resources, including hydrogeological and geological characteristics and the distribution of groundwater tables in space. Essential for water planning efforts, they help identify overused wells and places that need artificial groundwater replenishment. Vafadar et al. (2023) noted that GPMs also provide light on the dynamic interaction between groundwater reservoirs and human settlements.

Traditional methods such as drilling, geophysical, geological, and hydrogeological surveys for the prediction of groundwater resources have drawbacks, including being time-consuming and costly (El-Mezayen and El-Hamid 2023; Çelik et al. 2024). In recent years, studies on the GPMs have been conducted using spatial modeling with Geographic Information Systems (GIS) (Lee et al. 2020; Bilgilioğlu 2021; Bilgilioglu et al. 2022; Mandal et al. 2022; Borah and Deka, 2023). Water resource management and the delineation of possible groundwater zones are two areas that greatly benefit from GIS, a flexible tool for handling spatial data (Rahmati et al. 2016). Improving data visualization and simplifying the use of spatial data manipulation tools helps to save time and money when evaluating groundwater quality and improving groundwater resource management (Azuma et al. 2021).

Numerous studies have delved into the modeling and mapping groundwater potential, showcasing a variety of methodologies and approaches. Using GIS and the analytic hierarchy process (AHP), Rahmati et al. (2014) mapped the potential for groundwater in the Kurdistan area of Iran. According to their results, the AHP technique demonstrated encouraging prediction accuracy, with rainfall being the most crucial element and slope the least. Naghibi et al. (2020) explored the assessment of groundwater spring potential using Extreme Gradient Boosting (XGBoost) and parallel random forest (RF) algorithms based on digital elevation model (DEM)-derived factors. Their findings highlighted the high accuracy of similar RF and XGBoost with a linear booster for GPMs. Exploring groundwater potential mapping with GIS-based hybrid artificial intelligence algorithms was investigated by Phong et al. (2021). Results showed that the Modified Real AdaBoost (MRAB)—Functional Tree (FT) model outperformed other combinations regarding prediction accuracy, according to their study. Park and Kim (2021) used a novel ensemble tree-based method to assess the groundwater potential and tested its forecasting capabilities. Their results showed that XGBoost had

the best prediction accuracy compared to other algorithms. Azma et al. (2021) applied statistical modeling for spatial groundwater potential mapping using GIS techniques. They identified the electrical conductivity of soil as the most influential factor on groundwater electrical conductivity (GEC), with the multiple linear regression (MLR) model yielding superior results in predicting GEC. Rasool et al. (2022a, b) focused on mapping groundwater productivity potential in the provincial capital of Baluchistan, Pakistan, using machine learning algorithms. Among the six ML algorithms considered, XGBoost, RF, and artificial neural network (ANN) demonstrated exemplary performance. Al-Kindi and Janizadeh (2022) investigated using machine learning algorithms and hyperparameter tuning to map groundwater's potential in semi-arid ecosystems. Their study showed that Bayesian-XGBoost, grid search-XGBoost, random search-XGBoost, support vector machine (SVM), and XGBoost models achieved high levels of accuracy when assessed during the training phase. Choudhary et al. (2022) identified groundwater potential zones in India's upper Godavari sub-basin using bi-variate, Multi-Criteria Decision-Making (MCDM), and advanced machine learning algorithms. Their investigation found that the ensembled bi-variate and MCDM models outperformed advanced machine learning models in accuracy.

Estimating groundwater levels, GPM, vulnerability, and quality has grown more dependent on machine-learning algorithms in recent years (Tran et al. 2021; Masroor et al. 2023; Tegegne et al. 2024). Machine learning algorithms offer several advantages, including better accuracy, faster processing speed, higher robustness, and lower computational costs than traditional methods (Dargan et al. 2020; Sarker et al. 2021; Farahani et al. 2022). Machine learning algorithms can also identify data trends or patterns without human intervention, improving prediction performance with available data (Zhong et al. 2021). Machine learning algorithms may also learn from data without any background knowledge or physical behavior of the modeled event, which is a huge plus (Pham et al. 2019). XGBoost leverages Taylor expansion to better approximate the loss function and optimize the negative gradient more efficiently and accurately (Dong et al. 2022). It also handles the problem of missing/sparse data and is improved by cross-validation, early stop, and parallel processing (Chang et al. 2018).

Nevertheless, the effectiveness of a machine learning algorithm, such as XGBoost, is contingent upon the careful selection and optimization of its hyperparameters (Ghatasheh et al. 2022). Proper tuning of hyperparameters is crucial for achieving higher accuracy. Manual tuning is one method for adjusting hyperparameters, but it can be cumbersome and time-consuming, involving trial and error (Elgeldawi et al. 2021). Hyperparameter optimization can be another way to solve the problems described above, avoiding all the risks and a lot of time required in human tuning machine learning model parameters (Song et al. 2023). Hyperparameter optimization procedures have various benefits, including less time for parameter tuning and improvement in the performance of machine learning algorithms (Yang and Shami 2020). Hyperparameter optimization refers to the process settings of hyperparameters that can be found much more quickly in the data with excellent results and in the shortest possible time (Ali et al. 2023).

The primary purpose of this research is to elevate the accuracy of GPMs, providing comprehensive insights into the distribution, quantity, and quality of groundwater resources. This work tries to improve the XGBoost algorithm's performance in identifying areas prone to groundwater in Kazerun and Kooh-Chenar counties, Iran, by combining bio-inspired metaheuristic algorithms. The goal is to overcome the limits of conventional approaches. This improvement in spatial modeling contributes to more effective groundwater resource management. The innovation of this research lies in integrating advanced Geospatial Artificial Intelligence (GeoAI) techniques with bio-inspired metaheuristic algorithms. The application of invasive weed optimization (IWO) and biogeography-based optimization (BBO) to optimize the XGBoost algorithm represents a novel approach, addressing the challenges of hyperparameter tuning and improving modeling accuracy.

Workflow

The research for modeling and generating GPMs in Kazerun and Kooh-Chenar counties, Fars Province, was conducted in four stages, as depicted in Fig. 1. In the first step, all the dependent and independent data were collected and eventually used to construct a spatial database. The dependent variable is a spring inventory map of the catchment. Independent variables are as follows: slope, aspect, lithology, stream power index (SPI), normalized difference vegetation index (NDVI), topographic wetness index (TWI), distance from rivers, altitude, drainage density, rainfall, distance from faults, plan and profile curvature, slope length and steepness factor (LS-factor), land use/cover, and soil texture. In the second stage of the study, which involved choosing predictor variables for modeling, multicollinearity analysis was applied to detect and remove highly correlated factor variables. This analysis was conducted by computing the variance inflation factor (VIF) coefficients to eliminate predictor factors with high multicollinearity. After the factors were filtered through multicollinearity analysis, the frequency ratio (FR) method was employed to determine weights for the various levels of the criteria category. The FR values quantifying groundwater occurrence were computed based on the proportion of groundwater occurrence distributions

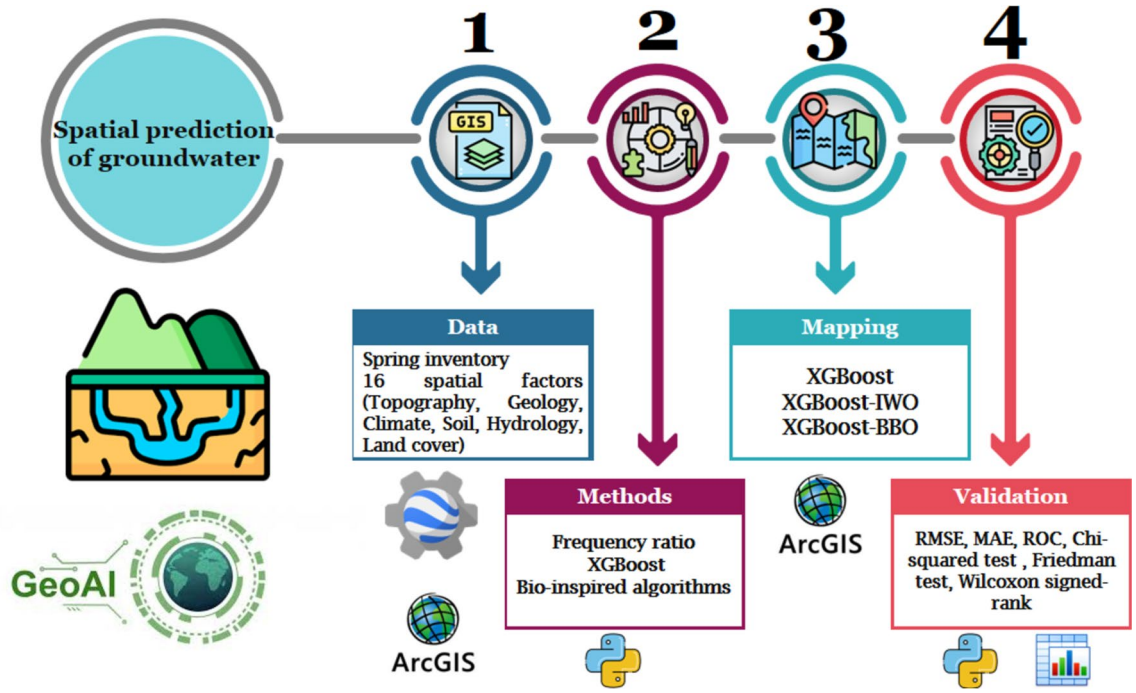


Fig. 1 Research framework stages

contained in each factor class and the total area of occurrences in the class. Then, the XGBoost algorithm was optimized using two metaheuristic algorithms, including BBO and IWO. In the third phase, GPTs were produced using the XGBoost, XGBoost-BBO, and XGBoost-IWO methods. In the last phase, results were evaluated by using the mean absolute error (MAE) and root mean square error (RMSE) indices, receiver operator characteristic (ROC) graphs, area under the curve (AUC) index, and, in the case study scope, the Chi-squared test, Friedman test, and Wilcoxon signed-rank test indices.

Material and methods

Case study

The Kazerun and Kooh-Chenar countries cover an area of 2,070 hectares and are situated southwest of Fars Province, Iran (Fig. 2). At the time of the 2016 census, it had a population of 266,217, making it the 53rd most populous city in the country. The county covers an area of approximately 4,119 km^2, and an average elevation of about 950 m above mean sea level. It lies between 29°6′ to 29°55′ north latitude and 51°16′ to 52°8′ east longitude. The region is characterized by moderately high mountainous terrain with an average annual temperature of 23 °C. A large part of the region (I.E., eastern, southeastern) is highly vegetated, including the heights of Mount Sorkhi and the Arg-een Plain. The mountainous area is home to many plains, the most important of which are Kazerun Plain, Jereh Plain, and Kamaraj Plain. Kazerun has a warm climate with rainy winters and dry summers. The average annual rainfall in the area is about 370 mm, making it one of the wettest places in Fars Province. Despite intensifying physical development and measures to increase agricultural and industrial activities, this area still struggles with rapid population growth. The extraction of groundwater, the most critical source of potable water and water used for irrigation and industry, has, in effect, reached abnormal levels that have raised concerns among stakeholders. With the exacerbation of drought periods, the static level of aquifers and other underground water resources, such as springs, wells, and qanats, has decreased. As the groundwater resources supply not only the water needed for agriculture but also a significant portion of the drinking water and water for industry, its evaluation in this region has become paramount, as various hazards threaten it.

Springs inventory

A spring represents a concentrated groundwater discharge on the Earth's surface, influenced by subsurface permeability and topography variations, leading to groundwater convergence (Bon et al. 2021). Spring flow magnitude and variability are contingent upon aquifer characteristics and groundwater recharge timing and volume. In this study, a

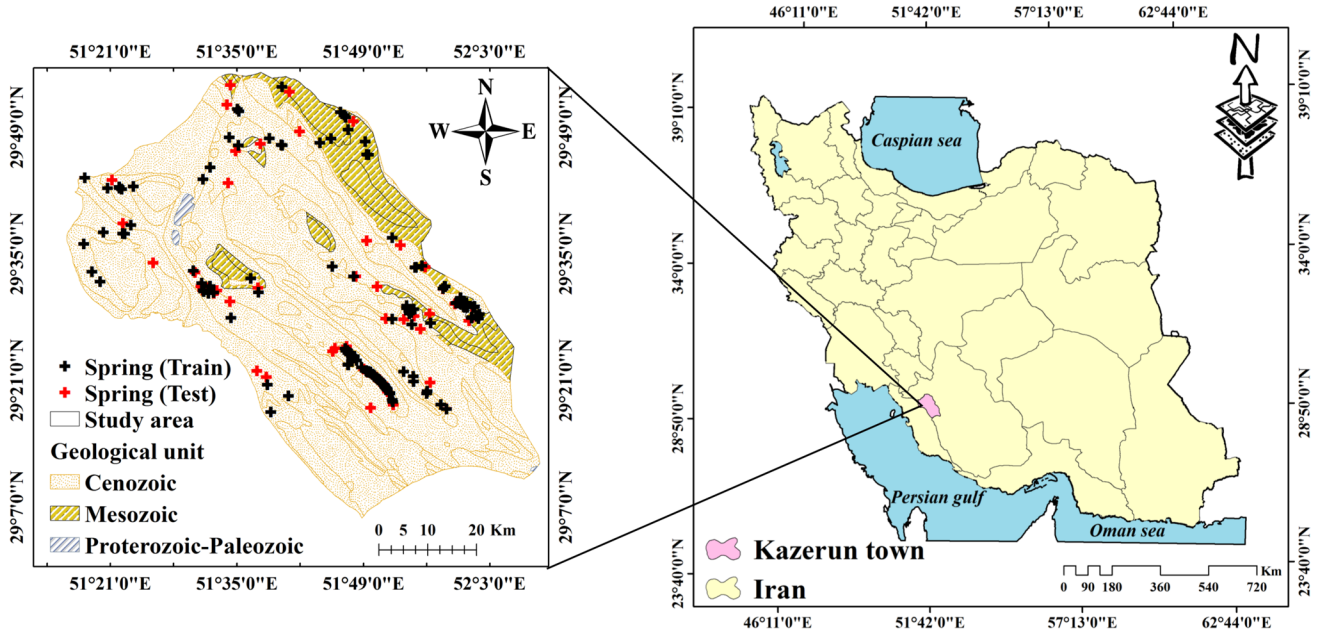


Fig. 2 Geographic location of the case study

spring inventory map is utilized as dependent data. The association between spring distribution and varied geoenvironmental parameters can be seen on the groundwater spring inventory map. This map shows the locations of springs in the research area, and it was provided by the Iranian Department of Water Resources Management (IDWRM). The next step was to conduct comprehensive field surveys using a Global Positioning System (GPS) to identify where these springs existed. For this purpose, 150 spring locations were considered in the modeling, with a value of one assigned as occurrence points, and 150 spring locations were randomly generated with a value of zero as non-occurrence points. In the next step, the Holdout method was employed, utilizing 70% of the data for modeling and 30% for evaluating the predicted map (Fig. 2).

Effective factors

In this study, effective criteria for groundwater potential were selected based on previous research and include topography, hydrology, climate, land properties, geology, and soil properties criteria (Khosravi et al. 2018; Chen et al. 2019; Al-Kindi et al. 2022; Masroor et al. 2023) (Fig. 3a-p and Table 1).

The DEM was created using the Shuttle Radar Topography Mission (SRTM) images in the GEE system. This retrieved topography-based criteria such as altitude, slope, aspect, plan curvature, profile curvature, TWI, SPI, and LS factor. Altitude is recognized as the unevenness of the land and, like slope, affects soil permeability, vegetation cover, and groundwater occurrence. At lower elevations, surface water recharge is high, and infiltration is increased, whereas runoff is rapid and copious at higher altitudes (Benjmel et al. 2022). Slope is a land feature that can help identify groundwater conditions by controlling infiltration (Saud 2010; Rahmati et al. 2014). Aspect indicates the slope direction and controls the physiographic trend, influencing the accumulation of surface water runoff (Masroor et al. 2023). Profile and plan curvature affect flow velocity variations, convergence and divergence on topographical surfaces, and other topographic features (Al-Abadi et al. 2016). To mitigate the effect of topography on runoff volume, Beven and Kirkby (1984) offered TWI as an improved model within the rainfall-runoff model. As shown in Eq. 1, it provides a physical indication of the distribution of soil moisture and the locations of surface-saturation zones (Costache 2019; Yariyan et al. 2022). Equation 2 determines the SPI, which assesses the extent to which slope erosion occurs due to flowing water (Park et al. 2017). The LS factor represents the amalgamation of slope steepness (S) and slope length (L), where their combined influence contributes to soil loss formation (Moore and Burch 1986; Moghaddam et al. 2015) (Eq. 3).

$$\mathrm{TWI} = \ln\left(\frac{\mathrm{Ac}}{\tan\beta}\right) \tag{1}$$

$$\mathrm{SPI} = \mathrm{Ac} \times \tan\beta \tag{2}$$

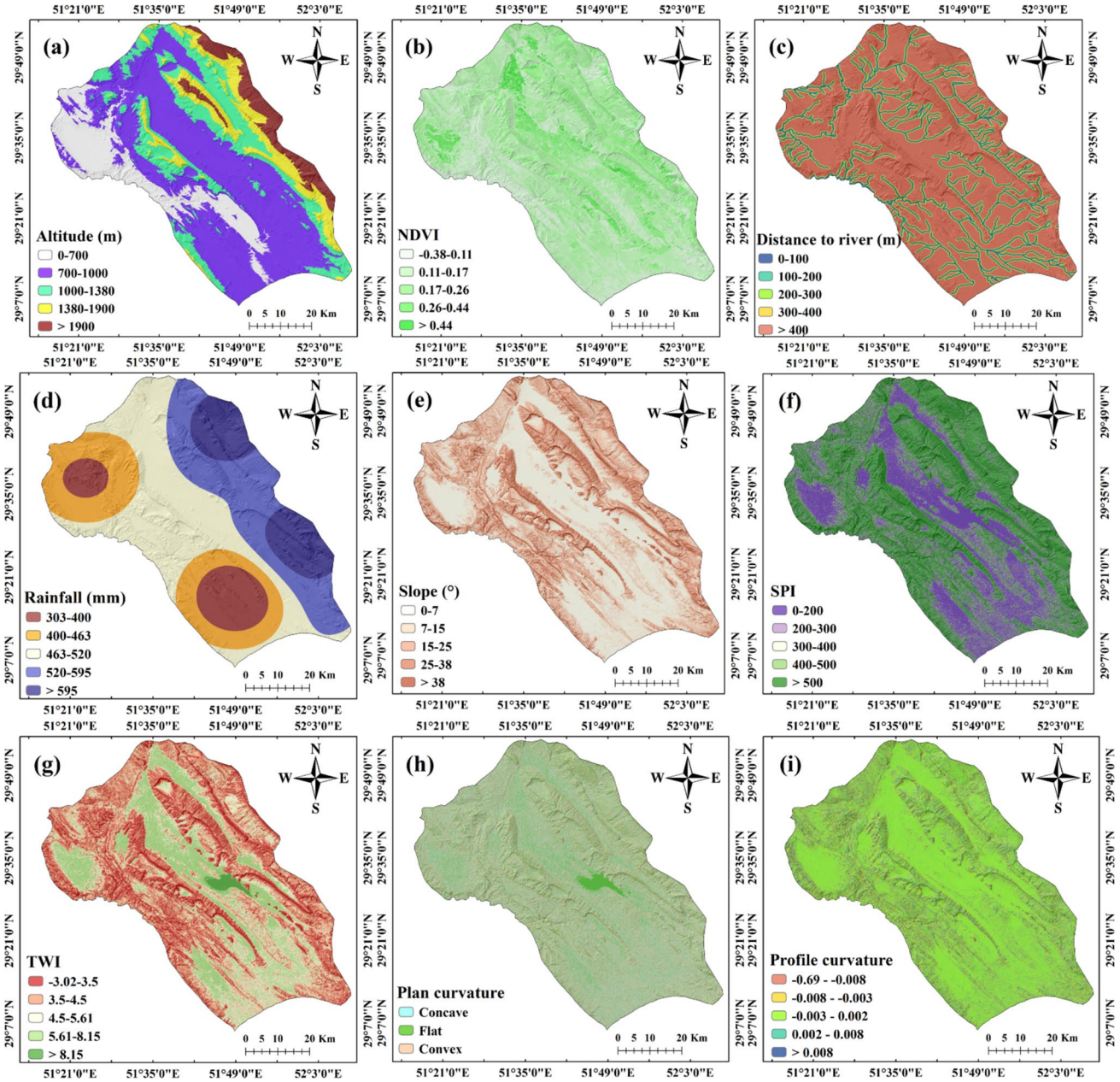


Fig. 3 Factors influencing groundwater: **a** Altitude, **b** NDVI, **c** Distance to river, **d** Rainfall, **e** Slope, **f** SPI, **g** TWI, **h** Plan curvature, **i** Profile curvature, **j** Drainage density, **k** Distance to fault, **l** LS factor, **m** Aspect, **n** Soil texture, **o** Land cover, and **p** Lithology

$$\mathrm{LS} = \left(\frac{\mathrm{Ac}}{22.13}\right)^{0.6}\left(\frac{\sin\beta}{0.0896}\right)^{1.3} \tag{3}$$

where Ac represents the upslope contributing area, and β is the slope. A 30 × 30 m pixel size was used to prepare and process criteria based on topography in the software packages ArcGIS 10.8 and SAGA GIS 2.1.2.

Land-based criteria, including land cover and NDVI, were obtained and processed using Landsat-8 images in the GEE system, with a pixel size of 30 × 30 m. Land cover reflects the intensity of human activities, which affects existing groundwater resources (Pham et al. 2019). The permeability of pastures and agricultural fields is higher than that of solid and impermeable residential and urban areas, which causes more runoff (Moghaddam et al. 2015). An RF algorithm was employed with 500 samples to generate the land cover map. In the end, six classes were produced: water bodies, agricultural areas, range land, bare land, forests, and urban areas. Changes in groundwater

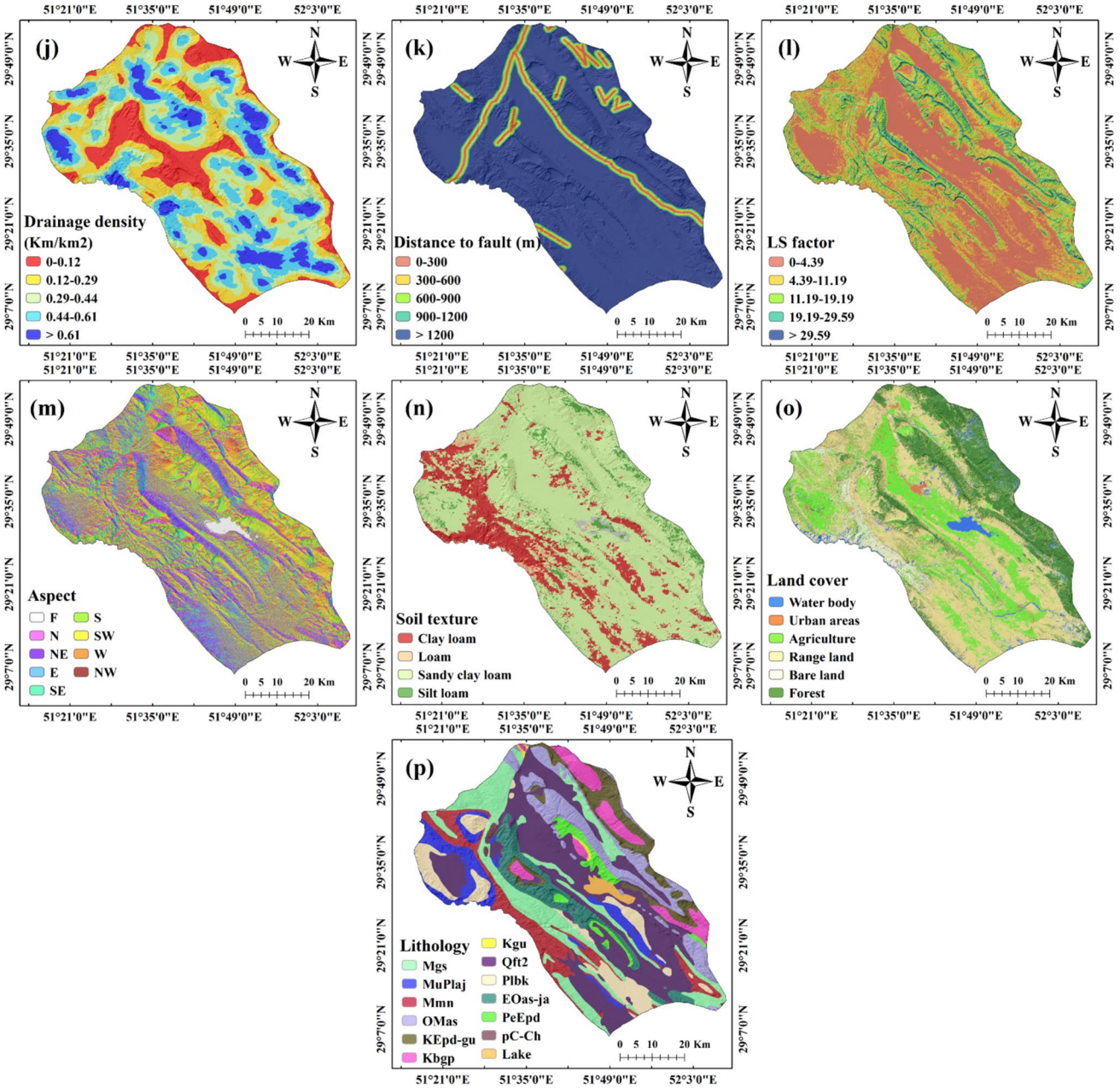


Fig. 3 (continued)

levels and groundwater flow are also associated with NDVI (Miraki et al. 2019). Generally, vegetation reflects the presence of water tables close to the surface (Benjmel et al. 2022; Farahani et al. 2023a) (Eq. 4).

$$\mathrm{NDVI} = \frac{(\mathrm{NIR} - \mathrm{Red})}{(\mathrm{NIR} + \mathrm{Red})} \tag{4}$$

where the NIR (Near-Infrared) band is subtracted from the red band, and the result is divided by the sum of the NIR and red bands. NDVI values typically range between − 1 and + 1, with higher values indicating denser vegetation cover and lower values corresponding to sparse vegetation or barren land (Saqr et al. 2022a).

Geology-based criteria, including lithology and faults, were prepared using the 1:100,000 geological map obtained from the Geological Survey of Iran. Lithologists describe the mineralogy and physical characteristics of various rock and soil units (Pham et al. 2019). The lithology map units were converted from polygons to raster format with a pixel size of 30 × 30 m using ArcGIS 10.8 software. The distance to faults is critical as faults provide opportunities for water

Table 1 Sources of preparation of spatial factors affecting groundwater

Factors	Data source	Scale/Resolution
Rainfall	Iranian Meteorological Organization (IMO)	30 m
Altitude	Shuttle radar topography mission (SRTM) images	30 m
Slope	Shuttle radar topography mission (SRTM) images	30 m
Aspect	Shuttle radar topography mission (SRTM) images	30 m
Plan curvature	Shuttle radar topography mission (SRTM) images	30 m
Profile curvature	Shuttle radar topography mission (SRTM) images	30 m
LS factor	Shuttle radar topography mission (SRTM) images	30 m
Topographic wetness index (TWI)	Shuttle radar topography mission (SRTM) images	30 m
Stream power index (SPI)	Shuttle radar topography mission (SRTM) images	30 m
Distance from the river	Shuttle radar topography mission (SRTM) images	30 m
Drainage density	Shuttle radar topography mission (SRTM) images	30 m
Lithology	Geological maps of Iran	1:100,000
Distance to fault	Geological maps of Iran	1:100,000
Land cover	Sentinel-1 and Sentinel-2 images	30 m
Soil type	Isfahan regional water company	1:100,000
Normalized difference vegetation index (NDVI)	Landsat-8 satellite imagery	30 m

infiltration beneath the surface (Yariyan et al. 2022). For the distance-to-fault map, Euclidean distance analysis with a pixel size of 30 × 30 m was utilized in ArcGIS 10.8.

In this study, two hydrology-based criteria, including distance from rivers and drainage density, were considered. The distance to rivers represents the proximity of each location in the study area to the nearest river, which directly influences groundwater recharge rates (Arabameri et al. 2021). Drainage density is a determining factor controlling groundwater occurrence (Hussein et al. 2017). The DEM of the research region was used to create the river map. Afterward, we analyzed density and distance using ArcGIS 10.8. Groundwater potential mapping relies heavily on rainfall data because it directly impacts groundwater replenishment (Moghaddam et al. 2020). The rainfall map was created using kriging in ArcGIS 10.8 with average yearly rainfall data from 26 synoptic stations in Fars province from 2012 to 2022. The amount of water that can percolate through sandy soils and other coarse-textured soils is much higher than that through clayey and silty soils (Yariyan et al. 2022). A soil texture map was created using the GEE platform, adhering to the soil classification system established by the USDA (U.S. Department of Agriculture).

Preprocessing techniques

Multicollinearity analysis

A multicollinearity analysis method was used in this study to investigate the relationships between factors that influence groundwater. Multicollinearity generally occurs when there is a significant correlation between the factors affecting a phenomenon (Kalnins 2018). In this regard, independent variables may exhibit significant correlations in addition to the relation between independent and dependent variables, significantly impacting the interpretation of the results (Farahani et al. 2022). Multicollinearity in multiple regression is quantified by using the VIF (Eq. 5) (Farahani et al. 2023b).

$$\mathrm{VIF} = \frac{1}{\mathrm{Tolerance}} = \frac{1}{1 - R^2} \tag{5}$$

where R^2 is the squared R-squared of the regression. If VIF > 10, it indicates the presence of collinearity among the input variables (Mirzaee et al. 2016; Shrestha 2020).

Frequency ratio (FR) method

The basic idea behind this technique is to assess the observed geographical association between springs and various factors contributing to groundwater potential (Falah and Zeinivand 2019). FR analysis calculates the ratio of groundwater occurrence likelihood to the probability of groundwater absence for each class of conditioning factors using Eq. 6 (Lee et al. 2020).

$$\mathrm{FR} = \frac{P_l}{P_c} \tag{6}$$

where P_c represents the percentage of pixels in each factor class, while P_l represents the percentage of groundwater pixels in each class.

Modeling techniques

In the context of selecting the XGBoost model's hyperparameters, which include learning rate, n_estimators,

max_depth, min_child_weight, gamma, subsample, and colsample_bytree, two bio-inspired optimization algorithms, IWO and BBO, were utilized. Therefore, both algorithms aimed at optimizing these hyperparameters of XGBoost iteratively over multiple iterations based on an objective function, i.e., minimization of RMSE (Eq. 7) (Saqr et al. 2022b).

$$\mathrm{RMSE} = \sqrt{\frac{\sum_{i=1}^{n}\left(y_i - \overline{y_i}\right)^2}{N}} \tag{7}$$

The RMSE represents the root mean square error, where y_i denotes the observed values, $\overline{y_i}$ is the mean of the predicted values, and N is the total number of observations. The algorithms then evolved the solutions through their respective mechanisms: IWO simulated weed growth through reproduction and dispersal, while BBO mimicked migration between different solution sets. The process continued until the RMSE stopped improving significantly, identifying the optimal hyperparameters for the XGBoost model. Custom stopping criteria and parallel processing were also implemented to enhance computational efficiency and ensure convergence.

Extreme Gradient Boosting (XGBoost) algorithm

Chen and Guestrin (2016) introduced the XGBoost algorithm to solve classification and regression tasks. It has gained significant popularity among researchers due to its high execution speed (Song et al. 2022). XGBoost combines numerous weak learners using additive learning to create a powerful learner (Rasool et al. 2022a, b). The advantages of XGBoost include parallel learning, high flexibility, internal cross-validation, etc., and it demonstrates good performance (Wu et al. 2021).

The basic idea behind the XGBoost method is to fit the residual predictions, learn new features repeatedly with the help of a tree structure, and then get the sample score. The final score for the sample prediction is derived by adding all the scores from all the trees (Song et al. 2022). XGBoost can prevent overfitting and provide more accurate predictions through regularized objective, shrinkage, and column subsampling (Lin et al. 2022).

The regularized objective term can allow for simple and efficient model selection. When the model is trained, column subsampling prevents overfitting as it requires less memory and is computationally more efficient than subsampling by rows (Lin et al. 2022). A more detailed explanation of the principles for the model XGBoost is described in the following sections (Jiang et al. 2020). The XGBoost algorithm optimizes the objective function in training the model using the training method through addition. The objective function for the t-th model of the model XGBoost can be expressed by Eq. 8 (Li et al. 2020):

$$obj^{(t)} = \sum_{i=1}^{n} l\left(y_i, \hat{y}_i^{t-1} + f_t\left(x_i\right)\right) + \Omega\left(f_t\right) + \text{constan } t \tag{8}$$

where Ω denotes the regularization term of the model, the constant is a constant term, and l is the loss term of the t-th round. The regularization term Ω is obtained through the following Equation (Li et al. 2020):

$$\Omega\left(f_t\right) = \gamma \cdot T_t + \lambda \frac{1}{2} \sum_{j=1}^{T} w_j^2 \tag{9}$$

In the above equation, γ and λ are both customization parameters. Larger values for these parameters lead to more superficial tree structures, effectively addressing overfitting issues. In this section, a second-order Taylor expansion is performed on the first equation above ($obj^{(t)}$) according to the following Equation (Jiang et al. 2020):

$$\mathrm{obj}^{(t)} = \sum_{i=1}^{n} \left[l\left(y_i, \hat{y}_i^{t-1}\right) + g_i f_t\left(x_i\right) + \frac{1}{2} h_i f_t^2\left(x_i\right) \right] + \Omega\left(f_t\right) + \text{constan } t \tag{10}$$

According to Jiang et al. (2020), the following Equations describe the first derivative g, and the second derivative h:

$$g_i = \partial_{\hat{y}_i^{t-1}} l\left(y_i, \hat{y}_i^{t-1}\right) \tag{11}$$

$$h_i = \partial^2_{\hat{y}_i^{t-1}} l\left(y_i, \hat{y}_i^{t-1}\right) \tag{12}$$

In this section, the expressions for $\Omega\left(f_t\right)$, g_i, and h_i are substituted into the final equation $\mathrm{obj}^{(t)}$, and then the derivatives are taken. Solutions can then be obtained using the following Equations (Jiang et al. 2020):

$$w_j^* = -\frac{\sum g_i}{\sum h_i + \lambda} \tag{13}$$

$$\mathrm{obj}^* = -\frac{1}{2} \sum_{j=1}^{T} \frac{\left(\sum g_i\right)^2}{\sum h_i + \lambda} + \gamma \cdot T \tag{14}$$

where obj^* represents the score of the loss function (where a lower score indicates a better tree structure), and w_j^* is the solution for the weights.

Biogeography-based optimization (BBO) algorithm

Recommending the BBO algorithm is based on Simon's work from 2008 to find global optimization (Simon 2008). The BBO literature review inspires its methodology derived from biogeography science (Rahmati and

Zandieh 2011). Based on the BBO literature, the algorithm in this study performs competitively for both unimodal and multimodal functions, adopting evolutionary operations inspired by species migration among islands, species emergence, and species extinction processes. It is claimed to significantly outperform previous population-based metaheuristic algorithms (Santosa and Safitri 2015). The BBO method is simple, flexible, and computationally efficient. It relies upon random, stochastic, and without objective function derivatives.

BBO's search procedure is conceptually comparable to other evolutionary algorithms, beginning by producing a collection of randomly selected solutions. The fitness function assesses the random solutions, and evolution occurs over several iterations (Ma et al. 2017). According to Saremi et al. (2014), the habitat suitability index (HSI) represents the fitness value linked to each habitat in BBO. Residents tend to migrate according to HSI, favoring movement from high-HSI habitats to low-HSI ones. Conversely, low HSI habitats are prone to immigration from high HSI areas, with the possibility of random habitat changes regardless of HSI values. These three rules allow ecosystems to boost each other's HSI and develop the first haphazard answers to a given issue (Mirjalili et al. 2014).

In this algorithm, each habitat includes three rates: immigration (λ_k), emigration (μ_k), and mutation (Saremi et al. 2014) (Eqs. 15 and 16).

$$\mu_k = \frac{E \times n}{N} \tag{15}$$

$$\lambda_k = \mathrm{I} \times \frac{1 - \mathrm{n}}{\mathrm{N}} \tag{16}$$

In this context, n and N stand for the total number of inhabitants and maximum number of inhabitants, respectively; E and I are the maximum rates of emigration and immigration, respectively. As the number of habitats increases, the likelihood of emigration decreases; the reverse is true for immigration. Equations 17 and 18 define the mutation rate of BBO, which is a function of the number of inhabitants (Saremi et al. 2014; Lim et al. 2016):

$$m_n = M \times \left(1 - \frac{p_n}{p_{\max}}\right) \tag{17}$$

$$p_{\max} = \operatorname{argmax}\left(p_n\right) \tag{18}$$

where n = 1, 2,…, N, the mutation probability of the n-th habitat is denoted by p_n, and M is the user-defined beginning value for mutation.

Invasive weed optimization (IWO) algorithm

The IWO algorithm was proposed by Mehrabian and Lucas in 2006. This algorithm is random and employs a derivative-free optimization technique that mimics the colonizing behavior of weeds (Panda et al. 2018). The IWO algorithm demonstrates the capability to achieve excellent performance in solving mathematical equations through an approach involving adaptation and randomization of a weed colony (Huang et al. 2019). Weeds, as robust plants with offensive growth habits, seriously threaten crops. A robust optimization tool can be derived from the characteristics of weed communities, which include resilience, adaptability, and unpredictability due to their excellent resistance and flexibility to environmental changes (Misaghi and Yaghoobi 2019; Ibrahim et al. 2022).

Features of this procedure that mimic weed behavior in their natural habitat are as follows (Misaghi and Yaghoobi 2019):

I. Primary population initialization: At the outset, a limited number of seeds are dispersed throughout the search space.

II. Depending on its fitness level, each seed grows into a flowering plant during the reproductive stage when it starts producing seeds (Eq. 19).

$$n\left(w_i\right) = \frac{S_{\max}\left(\max \text{ fit} - \text{fit}\left(w_i\right)\right) - S_{\min}\left(\text{fit}\left(w_i\right) - \min \text{ fit}\right)}{\max \text{ fit} - \min \text{ fit}} \tag{19}$$

where $S_{\max}$ and $S_{\min}$ represent the maximum number of seeds and the minimum number of seeds, respectively.

III. Spectral Spread: Step three involves using Eq. 20 to create a normal distribution for the group's seeds, which includes an average planting location and a standard deviation.

$$\sigma_t = \left(\frac{T - t}{T}\right)^n \left(\sigma_{\text{initial}} - \sigma_{\text{final}}\right) + \sigma_{\text{final}} \tag{20}$$

where T represents the maximum number of iterations, σ_t, σ_{initial}, and σ_{final} denote the current standard deviation, fundamental standard deviation, and the final standard deviation, respectively.

IV. Competitive Deprivation: Here, to keep the number of plants in the colony constant, the least fit grass is pulled out if the number of grasses exceeds the maximum.

All the weeds in the colony can participate in reproduction without mating, a crucial feature of this algorithm. More suitable plants can produce more seeds than less fortunate plants in the colony (Panda et al. 2018).

Validation techniques

This study used evaluation metrics, such as MAE, RMSE, ROC, AUC, Chi-squared, Friedman, and Wilcoxon tests. The chosen metrics complement each other by providing a well-rounded assessment of model performance. MAE and RMSE capture prediction errors, with RMSE being more sensitive to large errors. ROC and AUC evaluate classification accuracy, measuring the model's ability to distinguish between classes. The Chi-squared test checks feature independence, while the Friedman and Wilcoxon tests assess statistical significance in model comparisons, ensuring reliable performance across different datasets.

Validation of models

Evaluation of algorithm predictions using validation indices is an essential part of the modeling process that confirms the accuracy of predictions (Farhangi et al. 2021). In this study, two evaluation metrics, RMSE (Eq. 7) and MAE (Eq. 21), were employed to assess the performance of the developed models (Razavi-Termeh et al. 2023).

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}\left|y_i - \overline{y_i}\right| \tag{21}$$

The Equation states that y_i represents the actual value for each observation, and $\overline{y_i}$ stands for the predicted value. (Chicco et al. 2021).

Validation of potential maps

Various indices, such as ROC-AUC, Chi-square test, Friedman test, and Wilcoxon test, were employed to evaluate the GPM. According to Golkarian et al. (2018), diagnostic test models are often represented by ROC curves, and according to Bowers and Zhou (2019), AUC is a numerical metric that is connected with ROC curves that assess the model's capacity to distinguish across classes (Eq. 22). Values of the AUC statistic can fall between 0 and 1, while high values are related to better discrimination (Golkarian et al. 2018; Çelik et al. 2024). An AUC of 1 means that the model perfectly discriminates the positive from the negative cases with high specificity and sensitivity (Razavi-Termeh et al. 2022; Upwanshi et al. 2023; Paul and Roy, 2024).

$$\mathrm{AUC} = \int_{0}^{1} \mathrm{TPR}\ \mathrm{d(FPR)} \tag{22}$$

The true positive rate (TPR) and the false positive rate (FPR) are defined in this context, where n is the number of data points.

Chi-square testing ($\chi 2$) (Eq. 23) is a commonly used statistical approach for determining whether two variables are independent or associated. This test evaluates the independence of a class label from a specific feature in feature ranking (Sahin et al. 2016).

$$\chi^2 = \sum_{i=1}^{2}\sum_{j=1}^{k}\frac{\left(A_{ij} - E_{ij}\right)^2}{E_{ij}} \tag{23}$$

where k represents the quantity of classes, A_{ij} indicates the number of cases in the ith interval and jth class, and E_{ij} represents the anticipated probability.

The Friedman test is a statistical method to determine significant model differences (Ma et al. 2022). It is specifically designed for analyzing nonparametric data in a two-way analysis of variance. The Wilcoxon test compares the performance of two or more samples from the same population (Fallah-Zazuli et al. 2019). The Friedman test determines if two models can be distinguished statistically by assuming that the null hypothesis (H0) is valid (p-value < 0.05). The Wilcoxon test, on the other hand, compares two models side by side and calculates p- and z-values. According to Nhu et al. (2020), models are considered statistically different if the p-value is less than 0.05 and the z-value falls between $+1.96$ and -1.96.

Result

Preprocessing of input data

Meticulous data preprocessing steps were conducted before modeling the groundwater potential in the study areas. Initial steps included detecting and treating multicollinearity among the predictor factors based on the VIF index (Table 2)

Table 2 Results of multicollinearity testing for effective groundwater factors

Factors	VIF value
Altitude	1.7
Aspect	1.2
Distance to fault	1.05
Distance to river	1.1
Drainage density	1.2
Land cover	1.7
Lithology	1.3
LS	1.09
NDVI	2.05
Plan curvature	1.2
Profile curvature	1.2
Rainfall	1.4
Slope	2.2
Soil texture	1.2
SPI	1.5
TWI	1.3

using MedCalc Statistical Software. Per the analysis, the highest value for VIF emerged from the slope (2.2), while the low-end value emerged from the factor of distance to fault (1.05). Multicollinearity testing ensured all the predictors had VIF scores below 10, which is considered acceptable for multicollinearity. Therefore, it was concluded that all predictors were plausible for the modeling phase of the study.

Figure 4 shows the results of the subsequent investigation stage, which involved using the FR method to determine how different factor values affected the presence of groundwater. This method, executed within ArcGIS 10.8, assigns higher values to factors indicating a more substantial impact on groundwater incidence. Lithology emerged as a pivotal factor, with the class KEpd-gu exhibiting the highest impact on groundwater occurrence, characterized by a notable value of 3.49. Altitude and slope factors demonstrated significant influence, with classes 0–700 m and 0–7 degrees yielding values of 2.92 and 1.1, respectively. The NDVI factors showcased noteworthy outcomes. The class > 0.44 in NDVI, with a value of 12.16, exerted the most substantial influence on groundwater occurrence. Similarly, the distance to the river factor highlighted the class 0–100 m with a weight of 1.82 as the most impactful. Soil texture and land cover factors played a crucial role, where silt loam and agricultural land cover classes, with weights of 1.61 and 2.52, respectively, demonstrated significant influence on groundwater occurrence. The SPI and TWI factors indicated the classes > 400 and 4.5–5.61, with weights of 1.3 and 1.64, respectively, as particularly important. Furthermore, plan curvature and slope aspect factors exhibited notable influence, with plan curvature's flat classes and southwest slope aspect, carrying weights of 1.17 and 1.97, respectively, showing substantial impact. The distance to fault factor identified the class 600–900 m with a weight of 1.75 as having the most significant influence on groundwater occurrence. The LS factor and drainage density revealed that classes > 2.9 for LS and > 0.61 for drainage density, with weights of 2.13 and 1.79, respectively, played a pivotal role in groundwater occurrence. Additionally, in the profile curvature factor, the class range of -0.69 to -0.008, with a weight of 1.21, was identified as having the most significant importance.

Optimization and modeling

Following the preprocessing and preparation of data for modeling, the input data in Excel format was transferred to the Google Colab environment (https://colab.research.google.com/) for modeling using the Python programming language. The data were split into a 70:30 ratio for training and testing the modeling input. To determine the hyperparameters of the XGBoost algorithm, two metaheuristic algorithms with 100 epochs and a population size of 50 were employed. The objective was to optimize the XGBoost algorithm's hyperparameters using the BBO and IWO algorithms by minimizing the objective function. The parameters utilized in each algorithm are summarized in Table 3, determined through a trial-and-error approach.

The metaheuristic algorithms, based on the number of iterations they pass over, provide optimal values for the hyperparameters regarding the objective function. The BBO algorithm contributed to the construction of the model by having an objective function value of 0.46, which means that it was more accurate in optimizing the XGBoost algorithm compared to the IWO algorithm, which had an objective function value of 0.5. The optimized hyperparameters from both metaheuristic algorithms are shown in the output in Table 4.

After determining optimized hyperparameters, training and testing data were used for modeling and forecasting. We employed three models, namely, XGBoost, XGBoost-BBO, and XGBoost-IWO. Table 5 demonstrates modeling evaluation results for training and testing data by algorithm. The differences in performance noted in the modeling evaluation results for the three models (XGBoost, XGBoost-BBO, and XGBoost-IWO) were substantial. Using RMSE indicated that the XGBoost baseline model resulted in values of 0.165 on the training data and 0.424 on the testing data. When the IWO algorithm was introduced for hyperparameter optimization, the model significantly improved with RMSE values of 0.13 on the training data and 0.416 on the testing data. The XGBoost-BBO model increased predictive accuracy as the model had the lowest RMSE values of 0.114 on the training data and 0.39 on the testing data. Again, for MAE, the baseline XGBoost model resulted in MAE values of 0.121 on the training data and 0.295 on the testing data. Applying the IWO algorithm significantly reduced MAE to 0.087 on the training and 0.287 on the test sets. The XGBoost-BBO model outperformed both, achieving the lowest MAE values of 0.082 on the training and 0.28 on the test sets. The results demonstrate the effectiveness of the optimized XGBoost models using metaheuristic algorithms (BBO and IWO), showcasing improved performance metrics compared to the baseline XGBoost algorithm. Improved prediction accuracy and generalizability are shown by the XGBoost-IWO and XGBoost-BBO models, which show decreased RMSE and MAE values on both the test and training datasets.

The error histograms related to the training and testing predictions of the three models are presented in Fig. 5, and the results are summarized in Table 6. In terms of the mean errors, XGBoost has a negligible mean error of −2.6e-08 for the training and −0.032 for the testing set. In contrast, XGBoost-IWO has experienced marginally higher mean errors of −0.0001 for the training and −0.054 for the testing datasets. With an overall consistent to slightly increasing error by dataset, XGBoost-BBO shows a small positive mean error of

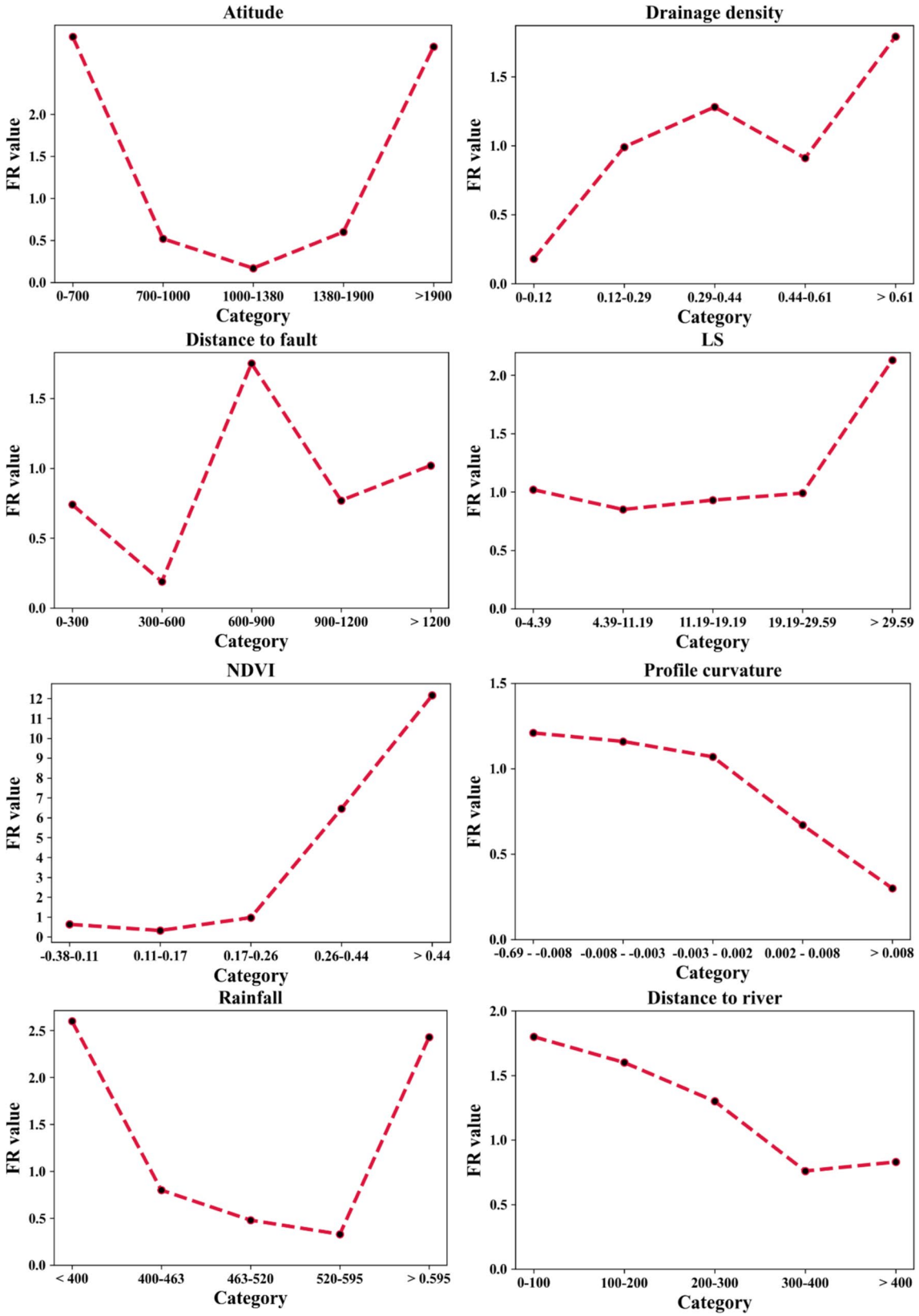


Fig. 4 Weight determination of factors influencing groundwater

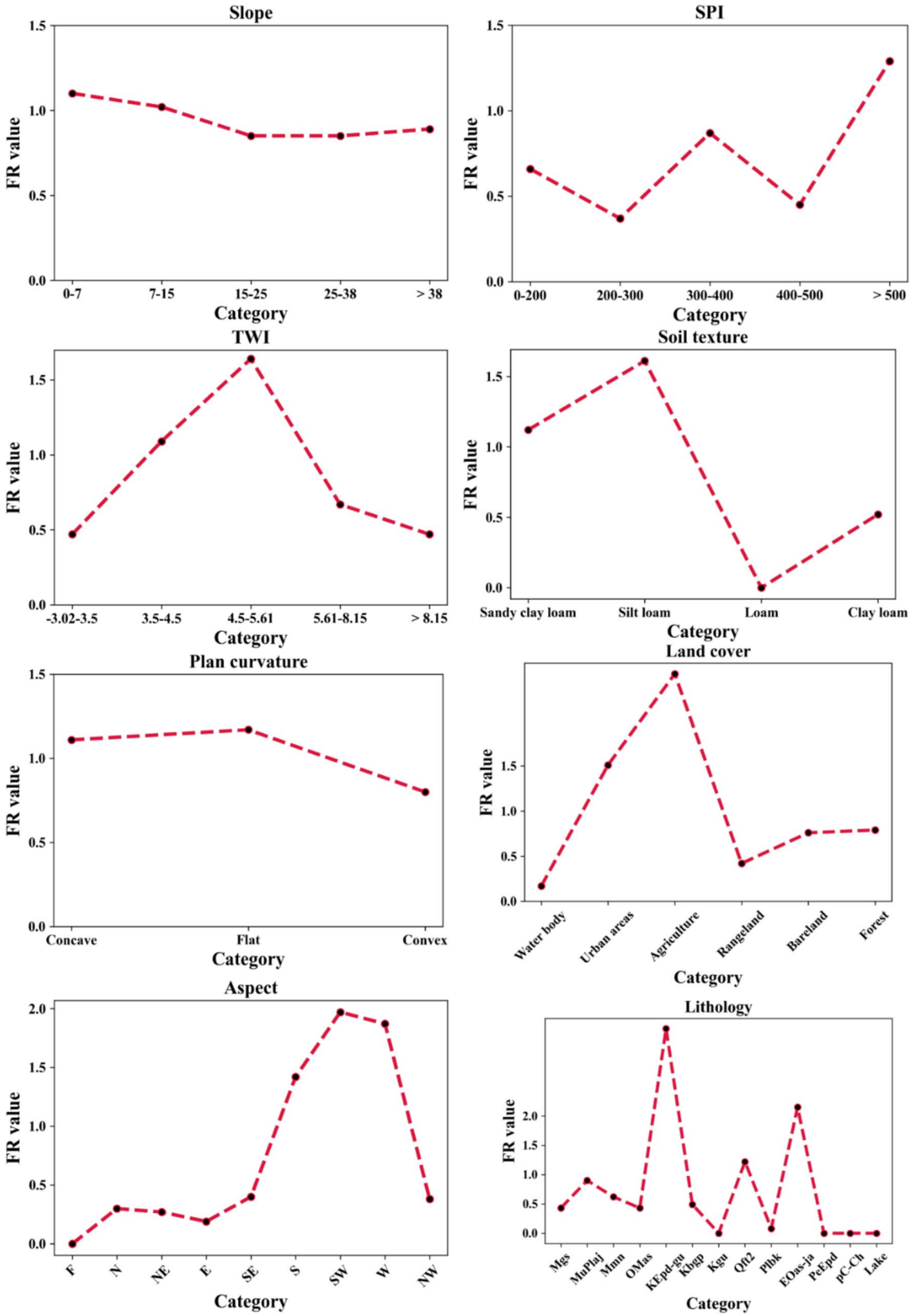


Fig. 4 (continued)

Table 3 Parameters used in bio-inspired metaheuristic algorithms

BBO algorithm	IWO algorithm
Epoch = 100	Epoch = 100
Population size = 50	Population size = 50
Mutation probability = 0.01	Number of Seeds (min) = 3
Number of elites will be keep for next generation = 2	Number of Seeds (max) = 9
	Variance Reduction Exponent = 3
	The initial value of Standard Deviation = 0.6
	The final value of Standard Deviation = 0.01

Table 4 Optimized hyper parameters for XGBoost algorithm

BBO algorithm	IWO algorithm
Learning rate = 0.1	Learning rate = 0.1
N estimators = 75	N estimators = 50
Max depth = 10	Max depth = 5
Min child weight = 10	Min child weight = 5
Gamma = 1e-05	Gamma = 0.01
Subsample = 1	Subsample = 1
Colsample bytree = 1	Colsample bytree = 1
Best cost = 0.46	Best cost = 0.5

Table 5 Evaluation results for developed GPM models

Models	RMSE		MAE	
	Train	Test	Train	Test
XGBoost	0.165	0.424	0.121	0.295
XGBoost-IWO	0.13	0.416	0.087	0.287
XGBoost-BBO	0.114	0.39	0.082	0.28

2.7e-05 for the training and −0.053 for the testing set. Regarding the standard deviation (STD) values, XGBoost shows an STD for the training set of 0.165, with a more significant increase to 0.42 in the testing set.

In contrast, XGBoost-IWO shows a lower STD of 0.111 for the training and 0.413 in the testing dataset. The XGBoost-BBO model also demonstrated similar trends in STD representation, with 0.120 STD for the training and 0.420 STD for the testing. In general, XGBoost and XGBoost-BBO approach similar mean errors, with XGBoost-BBO having a slightly positive mean error for the training dataset. XGBoost-IWO shows more significant mean errors across both training and testing sets. Finally, regarding variability, XGBoost-IWO has the smallest STD for both datasets, indicating a more consistent error pattern, while XGBoost and XGBoost-BBO exhibit a more varied set of errors.

Groundwater potential mapping (GPM) and validation

Three models were used to predict the groundwater potential throughout the study area. Using the Natural Breaks classification method, prediction maps were visually represented as five qualitative risk levels, ranging from very low to very high (Fig. 6). The summary of the mean and standard deviation for the prediction maps of groundwater potential is presented in Table 7. In the general study area, XGBoost-IWO showed the highest mean potential value (0.247), meaning that predictive mapping was generally elevated. XGBoost-BBO had the lowest mean value (0.01), meaning that prediction was low. Concerning variability, XGBoost-BBO showed the highest standard deviation (STD: 0.262), suggesting more spatial variation between the predictions. Conversely, XGBoost showed the lowest variability (STD: 0.235).

Table 8 displays the distribution of potential classes for groundwater in percentages for the three considered models. In the class "Very Low," the weaker class is held by XGBoost-BBO with 21.7%. On the contrary, XGBoost and XGBoost-IWO have higher percentages, with 25.8% and 26.15%, respectively. In the class "Low," XGBoost-BBO dominates with 31.76%, higher than XGBoost (31.7%) and XGBoost-IWO (42.13%). Models XGBoost, XGBoost-IWO, and XGBoost-BBO have 22.55%, 16.5%, and 27.44% in the class "Moderate." XGBoost-IWO has the worst result of the three models in the class "High" with 8.7%, where XGBoost (12.8%) and XGBoost-BBO (14.28%) have higher percentages. Ultimately, in the class "Very High," XGBoost-BBO has the lowest rate with 4.82%, and XGBoost (7.15%) and XGBoost-IWO (6.52%) presented higher percentages.

According to the evaluation results in MadCalc software for the GPM models using multiple gauges, including ROC, Chi-squared test, Friedman test, and the Wilcoxon signed-rank test. Table 9 and Fig. 7 illustrate the ROC: the AUC values for XGBoost, XGBoost-IWO, and XGBoost-BBO were 0.818, 0.831, and 0.837. The *P*-value for all models was less than 0.0001; the result was statistically significant. XGBoost-BBO presented the highest specificity rate at 0.8806, while XGBoost-IWO represented the most incredible sensitivity at 86.57%. The Youden Index J value was relatively the same between different models; XGBoost-BBO was 0.5672.

The Chi-squared test results (Table 10) confirmed statistical significance ($p < 0.0001$) for all models, with

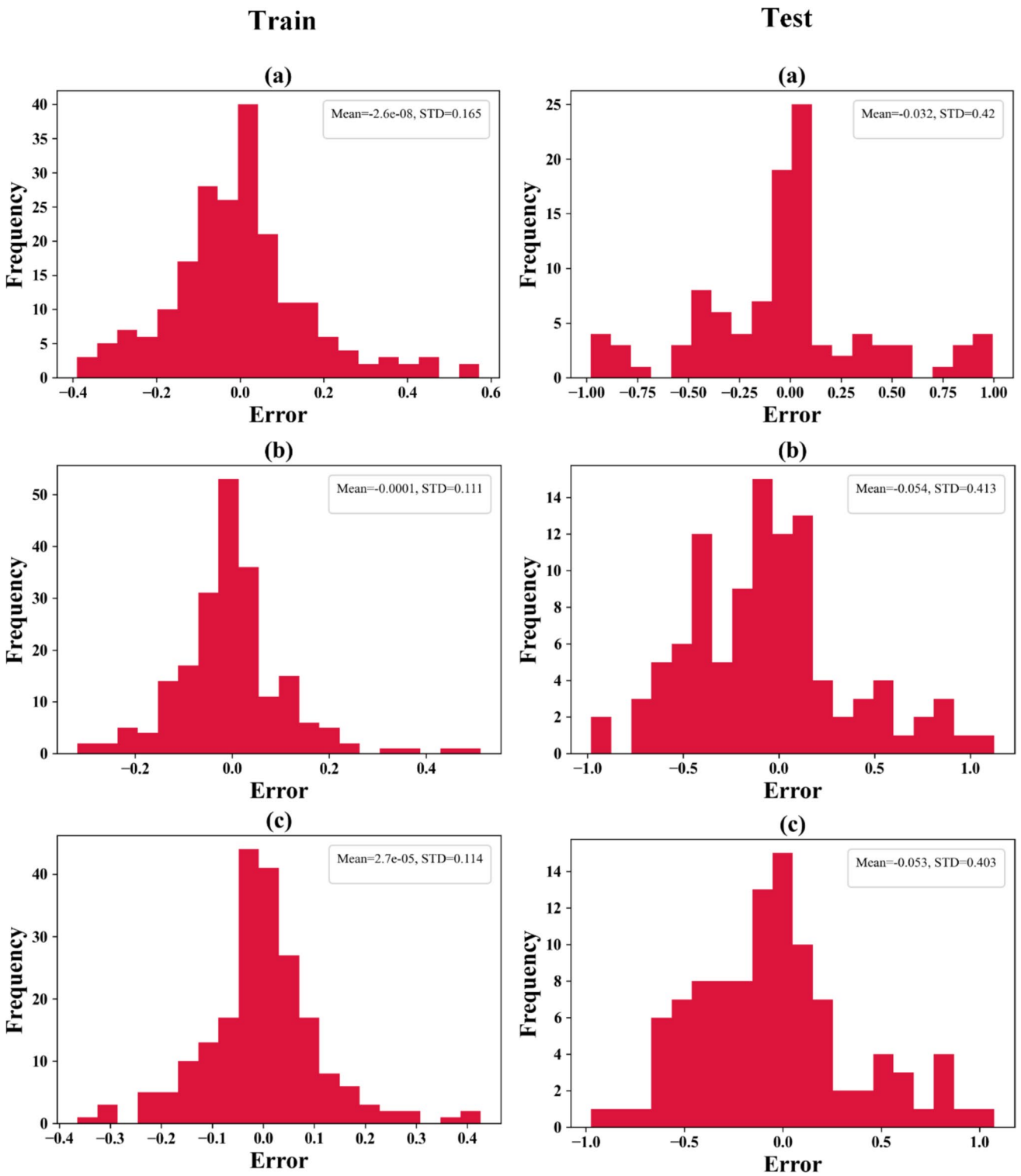


Fig. 5 Histogram error of GPM prediction models

Chi-squared values of 41.14, 44.09, and 45.69 for XGBoost, XGBoost-IWO, and XGBoost-BBO, respectively.

The Friedman test (Table 11) indicated a Friedman statistic of 91.38, suggesting a significant difference among the models. The mean ranks for XGBoost, XGBoost-IWO, and XGBoost-BBO were 1.87, 2.6, and 1.43, respectively.

Further investigation of the pairwise comparison using the Wilcoxon signed-rank statistics (Table 12) highlights that

Table 6 Mean and standard deviation of prediction errors

Models	Mean		STD	
	Train	Test	Train	Test
XGBoost	−2.6e-08	−0.032	0.165	0.42
XGBoost-IWO	−0.0001	−0.054	0.111	0.413
XGBoost-BBO	2.7e-05	−0.053	0.114	0.403

the models optimized via the metaheuristic approaches are superior. The comparison between XGBoost and XGBoost-IWO produced the Z statistic −8.002 ($p < 0.0001$) and favored XGBoost-IWO. Similarly, the comparison between XGBoost and XGBoost-BBO produced the Z statistic 6.30 ($p < 0.0001$) concerning XGBoost-BBO. XGBoost-IWO also performed better than XGBoost-BBO, evidenced by the Z statistic −8.6 ($p < 0.0001$).

The comprehensive evaluation results have revealed that the proposed metaheuristic-based models, such as XGBoost-IWO and XGBoost-BBO, have consistently performed better

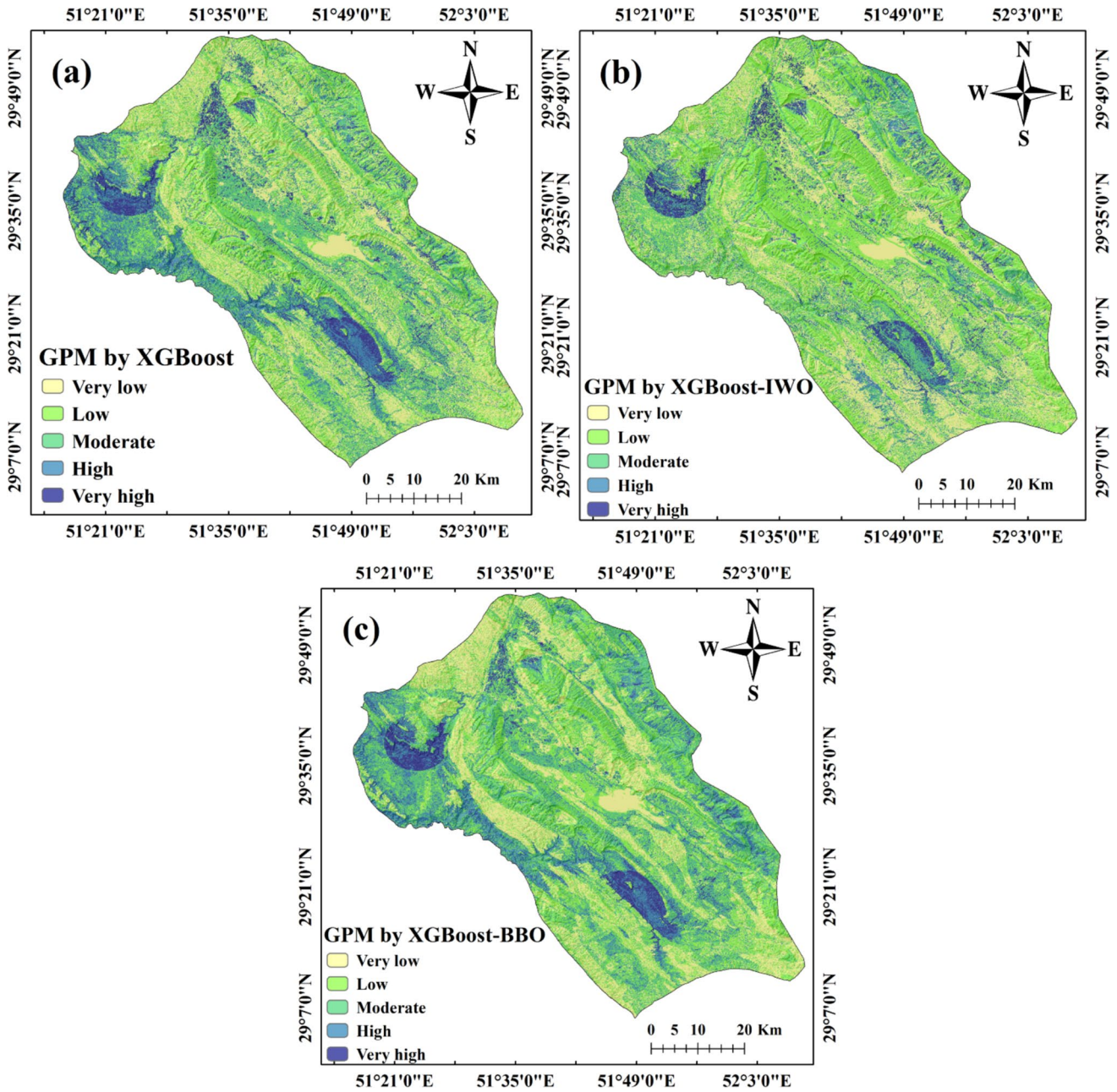


Fig. 6 Map of groundwater-prone areas

Table 7 Mean and standard deviation of prediction errors for groundwater-prone area maps

Models	Mean	STD
XGBoost	0.089	0.235
XGBoost-IWO	0.247	0.195
XGBoost-BBO	0.01	0.262

Table 8 Percentage of classes in groundwater-prone area maps

Potential class (%)	XGBoost	XGBoost-IWO	XGBoost-BBO
Very low	25.8	26.15	21.7
Low	31.7	42.13	31.76
Moderate	22.55	16.5	27.44
High	12.8	8.7	14.28
Very high	7.15	6.52	4.82

than the basic XGBoost model in the evaluation criteria such as ROC, Chi-squared test, Friedman test, and Wilcoxon signed-rank test. This will benefit the scientific community in terms of convenience in applying the metaheuristic for model development, thus making it easier for researchers to test and evaluate causal geography models more conveniently and with a higher confidence level.

Discussion

Comparison of spatial factors affecting groundwater

This study employed 16 influential factors affecting groundwater to predict potential groundwater zones. The FR approach was used to analyze the effects of each factor. Results of this approach revealed that, in the study area, the lithological unit KEpd-gu exerted the most significant influence on groundwater potential. The fine geological features of KEpd-gu, potentially related to permeability, porosity, or other hydrogeological properties, underscore its importance in shaping groundwater dynamics (Teimouri et al. 2018). The study further indicated that low altitude and gentle slope had a more pronounced effect on groundwater potential in the study area. Lower-altitude areas are often closer to rivers, and riverbanks and floodplains in low-altitude regions serve as natural conduits for feeding groundwater (Petitta et al. 2015). Low-slope terrain allows groundwater to accumulate and persist in subsurface layers (Sreedevi et al. 2005). Nearness to rivers also increased the likelihood of groundwater occurrence in the study area, as riverbanks act as direct access points to groundwater due to the pressure exerted by river flow (Su et al. 2017).

The study found that regions with more vegetation were more likely to have groundwater. Dense vegetation leads to improvements in soil structure, which in turn increases soil organic materials and water-holding capacity. As a result, enhanced soil moisture content recharged groundwater occurrence in the region (Lukenbach et al. 2019). The research indicates that the presence of agriculture and agricultural land cover greatly influenced the groundwater potential in the researched area. Farming activities can change soil structure and porosity, and both may help promote groundwater dynamics (Hussein et al. 2017). The study indicated a higher chance of groundwater occurring in areas with low SPI values. Low SPI values may signal areas where water flows are less erosive in water bodies, leading to less sediment transport and more likely to recharge the groundwater (Golkarian and Rahmati 2018). Groundwater occurrence in the research area was more affected by places with greater values of TWI. High TWI areas are often well-defined drainage networks connected to rivers, enhancing water exchange between surface water and groundwater reservoirs (Oh et al. 2011). Plan curvature demonstrated that areas with smooth surfaces had the highest impact on groundwater occurrence. Water in areas with plan curvature tends to remain in the subsurface for longer durations, contributing to prolonged groundwater residence times (Pourtaghi and Pourghasemi 2014).

In the slope aspect, the southwestern aspect holds more incredible promise for groundwater presence. This aspect indicates varied hydrological processes, like sunlight exposure, the distribution of rainfall, and evaporation rates, which all contribute to groundwater dynamics (Adiat et al. 2020). The study also found that areas near geological faults held more potential for groundwater presence. Faults are both hydraulic conduits for groundwater movement and an influential factor in permeability and pathways for fluid movement (Fronzi et al. 2021). The study also identified that areas with higher LS index values were more significant for groundwater presence. High channel density allows more efficient movement of water through the subsurface.

Additionally, the rapid conveyance of infiltrating water to the groundwater system accommodates steeper groundwater velocities (Doke et al. 2020). The study also identified

Table 9 Evaluation results for groundwater-prone area maps

Models	AUC	Standard error	95% confidence interval	z statistic	Significance level	Sensitivity	Specificity	Youden index J
XGBoost	0.818	0.0371	0.743 to 0.880	8.575	< 0.0001	64.18	91.04	0.5522
XGBoost-IWO	0.831	0.0351	0.757 to 0.891	9.433	< 0.0001	86.57	68.66	0.5522
XGBoost-BBO	0.837	0.0349	0.763 to 0.895	9.645	< 0.0001	68.66	88.06	0.5672

that areas with lower values of profile curvature were more significant for groundwater presence. A lower profile curvature indicates areas with flatter landforms and potentially controls numerous hydrogeological factors contributing to groundwater dynamics (Benjemel et al. 2022). In the soil texture factor, silty loam soil texture was more significant. This particular soil texture was predominantly significant for its medium texture. It is of great importance because silty loam is a combination of silty and loam qualities. This combination of textures generally results in good water retention and infiltration conditions, directly influencing groundwater dynamics (Yadav et al. 2023).

Comparison of GPM models

The study leveraged an enhanced XGBoost model by integrating two metaheuristic algorithms, IWO and BBO, to predict spatially groundwater-prone areas. An advantage of XGBoost is it handles missing data and large databases efficiently (Zhang et al. 2020). However, our XGBoost model was not optimized for hyperparameter choices, which affected our modeling accuracy performance (Ghatasheh et al. 2022). Comparing the GPM models, particularly the baseline model, XGBoost, with XGBoost-IWO and XGBoost-BBO shows that integrating bio-inspired algorithms for hyperparameter optimization significantly improved model performance. The RMSE, MAE, ROC index, Chi-squared test, and Friedman analysis indicate that the metaheuristic-tuned XGBoost models perform best.

During optimization, the BBO algorithm outperformed the IWO method when optimizing the XGBoost algorithm. According to Liang et al. (2022), the BBO algorithm employs mutation and migration operators for intensification and diversification. Throughout the iteration process of the BBO algorithm, the migration operator tends to enhance each solution in the population, while the mutation operator increases diversity among them (Zheng et al. 2019). These features contribute to the superiority of this algorithm over the IWO algorithm in the optimization process. Additionally, despite its advantages, such as simplicity and ease of understanding (Bai et al. 2013), the IWO algorithm has drawbacks like low-quality initial population and susceptibility to locally optimal solutions, leading to premature convergence (Pan et al. 2014). Due to these disadvantages, the accuracy of this algorithm in the optimization process was lower than that of the BBO algorithm. These two metaheuristic algorithms have been employed in various studies to develop GPM in conjunction with machine learning algorithms. For instance, BBO has been used in combination with support vector regression (SVR) (Al-Fugara et al. 2022) and the adaptive neuro-fuzzy inference system) ANFIS((Chen et al. 2019). Additionally, GPM development has combined IWO with machine learning algorithms like ANFIS (AlAyyash et al. 2023; Khosravi et al. 2018). These studies have confirmed the capabilities of these two metaheuristic algorithms in GPM modeling.

This research achieved an accuracy of 83.7% using the XGBoost-BBO model to identify groundwater-prone areas. Comparing these results with previous studies employing the XGBoost model shows consistently high performance. Guo et al. (2023) reported 86.6% accuracy in the Yinchuan Plain, while Halder et al. (2024a, b) achieved 95.7% accuracy in Bankura, West Bengal. Park and Kim (2021) recorded 81.8% accuracy in Busan Metropolitan City, and Rasool et al. (2022a, b) achieved 98.3% accuracy in Baluchistan, Pakistan. Naghibi et al. (2020) reported 84.5% accuracy in the Borujerd region of Iran, while Halder et al. (2024a, b)

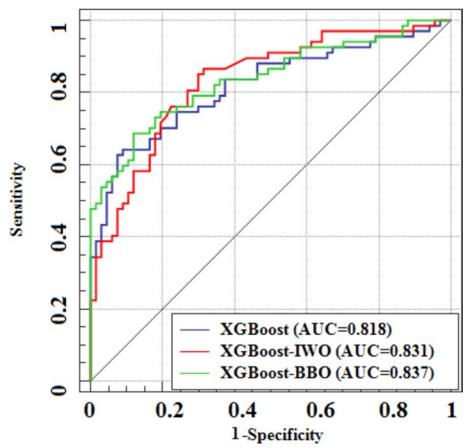


Fig. 7 Evaluation of groundwater-prone area maps with ROC curve

Table 10 Evaluation results for groundwater-prone area maps with Chi-squared test

Models	Chi-squared	Significance level ($P<0.0001$)
XGBoost	41.14	Yes
XGBoost-IWO	44.09	Yes
XGBoost-BBO	45.69	Yes

Table 11 Evaluation results for groundwater-prone area maps with Friedman test

Models	Mean rank	F value
XGBoost	1.87	
XGBoost-IWO	2.6	91.38
XGBoost-BBO	1.43	

Table 12 Evaluation results for groundwater-prone area maps with Wilcoxon signed-rank

	z statistic	Significance level
XGBoost ~ XGBoost-IWO	−8.002	$P < 0.0001$
XGBoost ~ XGBoost-BBO	6.3	$P < 0.0001$
XGBoost-IWO ~ XGBoost-BBO	−8.6	$P < 0.0001$

achieved 79% accuracy in the Shilabati watershed. Xiong et al. (2023) attained 89.9% accuracy in Luoning County, China, and Fatah et al. (2024) reported 88% accuracy in Iraqi Kurdistan. Overall, the findings indicate that the XGBoost model generally produces accuracies above 80%, demonstrating its effectiveness in modeling groundwater potential.

Limitations and future research directions

One of the study's drawbacks is the absence of analysis of water quality in locations with significant groundwater potential. In future research, beyond potential maps, researchers can incorporate water quality maps to better extract groundwater. Given that machine/deep learning algorithms have different structures in prediction, achieving a more accurate algorithm in prediction poses a challenging task, and examining all algorithms is time-consuming. Therefore, in future, researchers can provide a comprehensive framework of prediction algorithms for groundwater potential. Additionally, we consistently encounter various spatial factors and modeling uncertainties. To achieve higher accuracy, it is necessary to consider different uncertainties. Researchers can address this by incorporating multiple uncertainties into the modeling process.

Implications for groundwater management

The enhanced accuracy of the GPM will have real-world implications for groundwater management in the study area. With improved accuracy, these maps can more accurately define which zones are more likely to have groundwater resources, leading to better and more informed decisions regarding groundwater extraction, conservation, and sustainable use. Accurate GPMs can assist managers in optimizing well locations, avoid potential over-extraction in sensitive areas, and promote the management of groundwater resources more effectively. Moreover, enhancing accuracy in GPMs will assist in improving land-use planning policies, urban development planning, and other interventions that would protect and manage groundwater from an overall perspective of environmentally sustainable development.

Conclusion

This study aimed to propose a framework for enhancing the accuracy of groundwater potential mapping using the GeoAI approach. Metaheuristic algorithms with a biological background significantly improve the prediction of conventional XGBoost. Among all the developed models, the predictive accuracy of the XGBoost-BBO model exceeded that of other models for this specific study area. This suggests the potential role of combining biological metaheuristic algorithms with conventional machine learning algorithms. The spatial factors affecting most of the groundwater potential in this study area were low elevation and slope, high greenness, high drainage density, LS factor values, closeness to river or fault lines, higher rainfall, flat terrain, agricultural land, and high rainfall.

In conclusion, this research contributes to advancing predictive modeling for groundwater potential and underscores the significance of adopting a holistic GeoAI approach. The developed framework has practical implications for resource conservation, optimized groundwater utilization, and informed decision-making in water resource management. Future studies may delve deeper into refining the model and incorporating additional factors to understand groundwater dynamics in diverse geographical contexts comprehensively.

In summary, this study provides a platform for advanced forecasting of groundwater potential and underlines the need to emphasize a comprehensive GeoAI approach. The irrigation framework is of practical importance for resource management, efficient groundwater consumption, and water conservation-informed decisions. Future studies could concentrate on refining the model and introducing other factors that may play a role in understanding groundwater dynamics in different geographical settings. In summary, this study provides a platform for advanced forecasting of groundwater potential and underlines the need to emphasize a comprehensive GeoAI approach. The irrigation framework is of practical importance for resource management, efficient groundwater consumption, and water conservation-informed decisions. Future studies could concentrate on refining the model and introducing other factors that may play a role in understanding groundwater dynamics in different geographical settings.

Acknowledgements This work was supported in part by the ITRC Support Program under Grant IITP-2024-RS-2022-00156354 and in part by the Metaverse Support Program to Nurture the Best Talents under Grant IITP-2024-RS-2023-00254529 funded by the Ministry of Science and ICT of Korea and the Institute of Information and Communications Technology Planning and Evaluation (IITP).

Funding The author(s) received no specific funding for this work.

Data availability The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Conflict of interest The authors have no conflicts of interest to declare.

Ethical approval This article does not contain any studies with human participants or animals performed by any of the authors.

Informed consent Not applicable.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

References

Abd-Elmaboud ME, Saqr AM, El-Rawy M, Al-Arifi N, Ezzeldin R (2024) Evaluation of groundwater potential using ANN-based mountain gazelle optimization: a framework to achieve SDGs in East El Oweinat. Egypt J Hydrol Reg Stud 52:101703. https://doi.org/10.1016/j.ejrh.2024.101703

Aderemi BA, Olwal TO, Ndambuki JM, Rwanga SS (2021) A review of groundwater management models with a focus on IoT-based systems. Sustainability 14(1):148. https://doi.org/10.3390/su14010148

Adiat KAN, Ajayi OF, Akinlalu AA, Tijani IB (2020) Prediction of groundwater level in basement complex terrain using artificial neural network: a case of Ijebu-Jesa, southwestern Nigeria. Appl Water Sci. https://doi.org/10.1007/s13201-019-1094-6

Al-Abadi AM, Al-Temmeme AA, Al-Ghanimy MA (2016) A GIS-based combining of frequency ratio and index of entropy approaches for mapping groundwater availability zones at Badra–Al Al-Gharbi–Teeb areas, Iraq. Sustain Water Res Manag 2:265–283. https://doi.org/10.1007/s40899-016-0056-5

AlAyyash S, Al-Fugara A, Shatnawi R et al (2023) Combination of metaheuristic optimization algorithms and machine learning methods for groundwater potential mapping. Sustainability 15:2499. https://doi.org/10.3390/su15032499

Al-Fugara A, Ahmadlou M, Shatnawi R et al (2022) Novel hybrid models combining meta-heuristic algorithms with support vector regression (SVR) for groundwater potential mapping. Geocarto Int 37:2627–2646. https://doi.org/10.1080/10106049.2020.1831622

Ali Y, Awwad E, Al-Razgan M, Maarouf A (2023) Hyperparameter search for machine learning algorithms for optimizing the computational complexity. Processes 11:349. https://doi.org/10.3390/pr11020349

Al-Kindi KM, Janizadeh S (2022) Machine learning and hyperparameters algorithms for identifying groundwater Aflaj potential mapping in semi-arid ecosystems using LiDAR, Sentinel-2, GIS data, and analysis. Remote Sens 14:5425. https://doi.org/10.3390/rs14215425

Arabameri A, Pal SC, Rezaie F et al (2021) Modeling groundwater potential using novel GIS-based machine-learning ensemble techniques. J Hydrol Reg Stud 36:100848. https://doi.org/10.1016/j.ejrh.2021.100848

Azma A, Narreie E, Shojaaddini A et al (2021) Statistical modeling for spatial groundwater potential map based on GIS technique. Sustainability 13:3788. https://doi.org/10.3390/su13073788

Bai Y-Y, Xiao S, Liu C, Wang B-Z (2013) A hybrid IWO/PSO algorithm for pattern synthesis of conformal phased arrays. IEEE Trans Antennas Propag 61:2328–2332. https://doi.org/10.1109/tap.2012.2231936

Barzegar R, Razzagh S, Quilty J et al (2021) Improving GALDIT-based groundwater vulnerability predictive mapping using coupled resampling algorithms and machine learning models. J Hydrol 598:126370. https://doi.org/10.1016/j.jhydrol.2021.126370

Benjmel K, Amraoui F, Aydda A et al (2022) A multidisciplinary approach for groundwater potential mapping in a fractured semi-arid terrain (Kerdous Inlier, Western Anti-Atlas, Morocco). Water 14:1553. https://doi.org/10.3390/w14101553

Beven KJ, Kirkby MJ, Schofield N, Tagg AF (1984) Testing a physically-based flood forecasting model (TOPMODEL) for three U.K. catchments. J Hydrol 69:119–143. https://doi.org/10.1016/0022-1694(84)90159-8

Bilgilioğlu SS (2021) Land suitability assessment for Olive cultivation using GIS and multi-criteria decision-making in Mersin City Turkey. Arabian J Geosci. https://doi.org/10.1007/s12517-021-08768-8

Bilgilioglu SS, Gezgin C, Orhan O, Karakus P (2022) A GIS-based multi-criteria decision-making method for the selection of potential municipal solid waste disposal sites in Mersin Turkey. Environ Sci Pollut Res 29(4):5313–5329. https://doi.org/10.1007/s11356-021-15859-2

Bon AF, Mboudou GE, Banakeng AL, Ngoupayou JRN (2021) Piezometry-surface permeability statistical approach for identifying topographic zonation patterns of groundwater flows in the shallow weathered hard rock aquifer of the Olézoa watershed (Yaoundé–Cameroon). Environ Earth Sci. https://doi.org/10.1007/s12665-021-09874-2

Bowers AJ, Zhou X (2019) Receiver operating characteristic (ROC) area under the curve (AUC): a diagnostic measure for evaluating the accuracy of predictors of education outcomes. J Edu Stud Placed Risk (JESPAR) 24:20–46. https://doi.org/10.1080/10824669.2018.1523734

Çelik MÖ, Kuşak L, Yakar M (2024) Assessment of groundwater potential zones utilizing geographic information system-based analytical hierarchy process, vlse kriterijumska optimizacija kompromisno resenje, and technique for order preference by similarity to ideal solution methods: a case study in Mersin Türkiye. Sustainability 16(5):2202. https://doi.org/10.3390/su16052202

Chang Y-C, Chang K-H, Wu G-J (2018) Application of eXtreme gradient boosting trees in the construction of credit risk assessment models for financial institutions. Appl Soft Comput 73:914–920. https://doi.org/10.1016/j.asoc.2018.09.029

Chen W, Panahi M, Khosravi K et al (2019) Spatial prediction of groundwater potentiality using ANFIS ensembled with teaching-learning-based and biogeography-based optimization. J Hydrol 572:435–448. https://doi.org/10.1016/j.jhydrol.2019.03.013

Chicco D, Warrens MJ, Jurman G (2021) The coefficient of determination R-squared is more informative than SMAPE, MAE, MAPE, MSE and RMSE in regression analysis evaluation. PeerJ Comput Sci 7:e623. https://doi.org/10.7717/peerj-cs.623

Choudhary S, Pingale SM, Khare D (2022) Delineation of groundwater potential zones of upper Godavari sub-basin of India using bi-variate, MCDM and advanced machine learning algorithms. Geocarto Int 37:15063–15093. https://doi.org/10.1080/10106049.2022.2093992

Costache R (2019) Flash-flood potential assessment in the upper and middle sector of Prahova river catchment (Romania). A comparative approach between four hybrid models. Sci Total Environ 659:1115–1134. https://doi.org/10.1016/j.scitotenv.2018.12.397

Dargan S, Kumar M, Ayyagari MR, Kumar G (2020) A survey of deep learning and its applications: a new paradigm to machine learning. Arch Comput Methods Eng 27(4):1071–1092. https://doi.org/10.1007/s11831-019-09344-w

Doke A, Pardeshi SD, Das S (2020) Drainage morphometry and groundwater potential mapping: application of geoinformatics with frequency ratio and influencing factor approaches. Environ Earth Sci. https://doi.org/10.1007/s12665-020-09137-6

Dong J, Chen Y, Yao B et al (2022) A neural network boosting regression model based on XGBoost. Appl Soft Comput 125:109067. https://doi.org/10.1016/j.asoc.2022.109067

Elgeldawi E, Sayed A, Galal AR, Zaki AM (2021) Hyperparameter tuning for machine learning algorithms used for arabic sentiment analysis. Informatics 8:79. https://doi.org/10.3390/informatics8040079

El-Mezayen MM, El-Hamid HTA (2023) Assessment of water quality and modeling trophic level of lake Manzala, egypt using remotely sensed observations after recent enhancement project. J Indian Soc Remote Sens 51(1):197–211. https://doi.org/10.1007/s12524-022-01635-3

Falah NF, Zeinivand NH (2019) GIS-based groundwater potential mapping in Khorramabad in Lorestan, Iran, using frequency ratio (FR) and weights of evidence (WoE) models. Water Resour 46:679–692. https://doi.org/10.1134/s0097807819050051

Fallah-Zazuli M, Vafaeinejad A, Alesheykh AA et al (2019) Mapping landslide susceptibility in the Zagros Mountains, Iran: a comparative study of different data mining models. Earth Sci Inf 12:615–628. https://doi.org/10.1007/s12145-019-00389-w

Farahani M, Razavi-Termeh SV, Sadeghi-Niaraki A (2022) A spatially based machine learning algorithm for potential mapping of the hearing senses in an urban environment. Sustain Cities Soc 80:103675. https://doi.org/10.1016/j.scs.2022.103675

Farahani M, Razavi-Termeh SV, Sadeghi-Niaraki A, Choi S-M (2023a) A hybridization of spatial modeling and deep learning for people's visual perception of urban landscapes. Sustainability 15:10403. https://doi.org/10.3390/su151310403

Farahani M, Razavi-Termeh SV, Sadeghi-Niaraki A, Choi S-M (2023b) People's olfactory perception potential mapping using a machine learning algorithm: a Spatio-temporal approach. Sustain Cities Soc 93:104472. https://doi.org/10.1016/j.scs.2023.104472

Farhangi F, Sadeghi-Niaraki A, Razavi-Termeh SV, Choi S-M (2021) Evaluation of tree-based machine learning algorithms for accident risk mapping caused by driver lack of alertness at a national scale. Sustainability 13:10239. https://doi.org/10.3390/su131810239

Fatah KK, Mustafa YT, Hassan IO (2024) Groundwater potential mapping in arid and semi-arid regions of Kurdistan region of Iraq: a geoinformatics-based machine learning approach. Groundwater Sustain Dev 27:101337. https://doi.org/10.1016/j.gsd.2024.101337

Fronzi D, Mirabella F, Cardellini C et al (2021) The role of faults in groundwater circulation before and after seismic events: insights from tracers. Water Isotopes Geochem Water 13:1499. https://doi.org/10.3390/w13111499

Ghatasheh N, Altaharwa I, Aldebei K (2022) Modified genetic algorithm for feature selection and hyper parameter optimization: case of XGBoost in Spam prediction. IEEE Access 10:84365–84383. https://doi.org/10.1109/access.2022.3196905

Golkarian A, Rahmati O (2018) Use of a maximum entropy model to identify the key factors that influence groundwater availability on the Gonabad Plain Iran. Environ Earth Sci. https://doi.org/10.1007/s12665-018-7551-y

Golkarian A, Naghibi SA, Kalantar B, Pradhan B (2018) Groundwater potential mapping using C5.0, random forest, and multivariate adaptive regression spline models in GIS. Environ Monit Assess. https://doi.org/10.1007/s10661-018-6507-8

Gorelick SM, Zheng C (2015) Global change and the groundwater management challenge. Water Resour Res 51:3031–3051. https://doi.org/10.1002/2014wr016825

Guo X, Gui X, Xiong H, Hu X, Li Y, Cui H, Qiu Y, Ma C (2023) Critical role of climate factors for groundwater potential mapping in arid regions: Insights from random forest, XGBoost, and LightGBM algorithms. J Hydrol 621:129599. https://doi.org/10.1016/j.jhydrol.2023.129599

Halder K, Srivastava AK, Ghosh A, Nabik R, Pan S, Chatterjee U, Bisai D, Pal SC, Zeng W, Ewert F, Gaiser T, Pande CB, Islam ARMdT, Alam E, Islam MK (2024a) Application of bagging and boosting ensemble machine learning techniques for groundwater potential mapping in a drought-prone agriculture region of eastern India. Environ Sci Europe. https://doi.org/10.1186/s12302-024-00981-y

Halder S, Karmakar S, Maiti P, Roy MB, Roy PK (2024b) Application of machine learning and Fuzzy AHP for identification of suitable groundwater potential zones using field based hydrogeophysical and soil hydraulic factors in a complex hydrogeological terrain. Groundwater Sustain Dev 27:101329. https://doi.org/10.1016/j.gsd.2024.101329

Huang L, Asteris PG, Koopialipoor M et al (2019) Invasive weed optimization technique-based ANN to the prediction of rock tensile strength. Appl Sci 9:5372. https://doi.org/10.3390/app9245372

Hussein A-A, Govindu V, Nigusse AGM (2017) Evaluation of groundwater potential using geospatial techniques. Appl Water Sci 7:2447–2461. https://doi.org/10.1007/s13201-016-0433-0

Ibrahim A, Anayi F, Packianather M, Alomari OA (2022) New hybrid invasive weed optimization and machine learning approach for fault detection. Energies 15:1488. https://doi.org/10.3390/en15041488

Ifediegwu SI (2021) Assessment of groundwater potential zones using GIS and AHP techniques: a case study of the Lafia district Nasarawa State, Nigeria. Appl Water Sci. https://doi.org/10.1007/s13201-021-01556-5

Jiang H, He Z, Ye G, Zhang H (2020) Network intrusion detection based on PSO-Xgboost model. IEEE Access 8:58392–58401. https://doi.org/10.1109/access.2020.2982418

Kalnins A (2018) Multicollinearity: how common factors cause Type 1 errors in multivariate regression. Strateg Manag J 39:2362–2385. https://doi.org/10.1002/smj.2783

Kayastha V, Patel J, Kathrani N et al (2022) New Insights in factors affecting ground water quality with focus on health risk assessment and remediation techniques. Environ Res 212:113171. https://doi.org/10.1016/j.envres.2022.113171

Khosravi K, Panahi M, Bui DT (2018) Spatial prediction of groundwater spring potential mapping based on an adaptive neuro-fuzzy inference system and metaheuristic optimization. Hydrol Earth Syst Sci 22:4771–4792. https://doi.org/10.5194/hess-22-4771-2018

Lee S, Hyun Y, Lee S, Lee M-J (2020) Groundwater potential mapping using remote sensing and GIS-based machine learning techniques. Remote Sensing 12:1200. https://doi.org/10.3390/rs12071200

Li N, Li B, Gao L (2020) Transient stability assessment of power system based on XGBoost and factorization machine. IEEE Access 8:28403–28414. https://doi.org/10.1109/access.2020.2969446

Liang S, Fang Z, Sun G, Qu G (2022) Biogeography-based optimization with adaptive migration and adaptive mutation with its application in sidelobe reduction of antenna arrays. Appl Soft Comput 121:108772. https://doi.org/10.1016/j.asoc.2022.108772

Lim WL, Wibowo A, Desa MI, Haron H (2016) A Biogeography-based optimization algorithm hybridized with tabu search for the quadratic assignment problem. Comput Intell Neurosci 2016:1–12. https://doi.org/10.1155/2016/5803893

Lin L, Liang Y, Liu L et al (2022) Estimating PM2.5 concentrations using the machine learning RF-XGBoost model in Guanzhong Urban agglomeration China. Remote Sensing 14:5239. https://doi.org/10.3390/rs14205239

Lukenbach MC, Spencer CJ, Mendoza CA et al (2019) Evaluating how landform design and soil covers influence groundwater recharge in a reclaimed watershed. Water Resour Res 55:6464–6481. https://doi.org/10.1029/2018wr024298

Ma H, Simon D, Siarry P et al (2017) Biogeography-based optimization: a 10-year review. IEEE Trans Emerg Topics Comput Intell 1:391–407. https://doi.org/10.1109/tetci.2017.2739124

Ma J, Xia D, Guo H et al (2022) Metaheuristic-based support vector regression for landslide displacement prediction: a comparative study. Landslides 19:2489–2511. https://doi.org/10.1007/s10346-022-01923-6

Machiwal D, Jha MK, Mal BC (2011) Assessment of groundwater potential in a semi-arid region of india using remote sensing, GIS and MCDM techniques. Water Resour Manage 25:1359–1386. https://doi.org/10.1007/s11269-010-9749-y

Mandal T, Saha S, Das J, Sarkar A (2022) Groundwater depletion susceptibility zonation using TOPSIS model in Bhagirathi river basin, India. Model Earth Syst Environ 8(2):1711–1731. https://doi.org/10.1007/s40808-021-01176-7

Masroor M, Sajjad H, Kumar P et al (2023) Novel ensemble machine learning modeling approach for groundwater potential mapping in Parbhani District of Maharashtra. India Water 15:419. https://doi.org/10.3390/w15030419

Miraki S, Zanganeh SH, Chapi K et al (2019) Mapping groundwater potential using a novel hybrid intelligence approach. Water Resour Manage 33:281–302. https://doi.org/10.1007/s11269-018-2102-6

Mirjalili S, Mirjalili SM, Lewis A (2014) Let a biogeography-based optimizer train your multi-layer perceptron. Inf Sci 269:188–209. https://doi.org/10.1016/j.ins.2014.01.038

Mirzaee S, Ghorbani-Dashtaki S, Mohammadi J et al (2016) Spatial variability of soil organic matter using remote sensing data. CATENA 145:118–127. https://doi.org/10.1016/j.catena.2016.05.023

Misaghi M, Yaghoobi M (2019) Improved invasive weed optimization algorithm (IWO) based on chaos theory for optimal design of PID controller. J Comput Design Eng 6:284–295. https://doi.org/10.1016/j.jcde.2019.01.001

Moghaddam DD, Rezaei M, Pourghasemi HR et al (2015) Groundwater spring potential mapping using bivariate statistical model and GIS in the Taleghan Watershed. Iran Arabian J Geosci 8:913–929. https://doi.org/10.1007/s12517-013-1161-5

Moghaddam DD, Rahmati O, Panahi M et al (2020) The effect of sample size on different machine learning models for groundwater potential mapping in mountain bedrock aquifers. CATENA 187:104421. https://doi.org/10.1016/j.catena.2019.104421

Moore ID, Burch GJ (1986) Sediment transport capacity of sheet and rill flow: application of unit stream power theory. Water Resour Res 22:1350–1360. https://doi.org/10.1029/wr022i008p01350

Naghibi SA, Hashemi H, Berndtsson R, Lee S (2020) Application of extreme gradient boosting and parallel random forest algorithms for assessing groundwater spring potential using DEM-derived factors. J Hydrol 589:125197. https://doi.org/10.1016/j.jhydrol.2020.125197

Nguyen PT, Ha DH, Avand M et al (2020) Soft computing ensemble models based on logistic regression for groundwater potential mapping. Appl Sci 10:2469. https://doi.org/10.3390/app10072469

Nhu V-H, Mohammadi A, Shahabi H et al (2020) Landslide susceptibility mapping using machine learning algorithms and remote sensing data in a tropical environment. Int J Environ Res Public Health/Int J Environ Res Public Health 17:4933. https://doi.org/10.3390/ijerph17144933

Nobre RCM, Filho OCR, Mansur WJ et al (2007) Groundwater vulnerability and risk mapping using GIS, modeling and a fuzzy logic tool. J Contam Hydrol 94:277–292. https://doi.org/10.1016/j.jconhyd.2007.07.008

Oh H-J, Kim Y-S, Choi J-K et al (2011) GIS mapping of regional probabilistic groundwater potential in the area of Pohang City, Korea. J Hydrol 399:158–172. https://doi.org/10.1016/j.jhydrol.2010.12.027

Pan G, Li K, Ouyang A et al (2014) A hybrid clustering algorithm combining cloud model Iwo AND K-means. Int J Pattern Recognit Artif Intell 28:1450015. https://doi.org/10.1142/s0218001414500153

Panda DPK, Dutta S, Pradhan SK (2018) Optimal path planning for mobile robots using oppositional invasive weed optimization. Comput Intell 34:1072–1100. https://doi.org/10.1111/coin.12166

Park S, Kim J (2021) The predictive capability of a novel Ensemble Tree-Based algorithm for assessing groundwater potential. Sustainability 13(5):2459. https://doi.org/10.3390/su13052459

Park S, Hamm S-Y, Jeon H-T, Kim J (2017) Evaluation of logistic regression and multivariate adaptive regression spline models for groundwater potential mapping using R and GIS. Sustainability 9:1157. https://doi.org/10.3390/su9071157

Patra S, Mishra P, Mahapatra SC (2018) Delineation of groundwater potential zone for sustainable development: A case study from Ganga Alluvial Plain covering Hooghly district of India using remote sensing, geographic information system and analytic hierarchy process. J Clean Prod 172:2485–2502. https://doi.org/10.1016/j.jclepro.2017.11.161

Paul S, Roy D (2024) Geospatial modeling and analysis of groundwater stress-prone areas using GIS-based TOPSIS, VIKOR, and EDAS techniques in Murshidabad district. India Model Earth Syst Environ 10(1):121–141. https://doi.org/10.1007/s40808-022-01589-y

Petitta M, Caschetto M, Galassi DMP, Aravena R (2015) Dual-flow in karst aquifers toward a steady discharge spring (Presciano, Central Italy): influences on a subsurface groundwater dependent ecosystem and on changes related to post-earthquake hydrodynamics. Environ Earth Sci 73:2609–2625. https://doi.org/10.1007/s12665-014-3440-1

Pham BT, Jaafari A, Prakash I et al (2019) Hybrid computational intelligence models for groundwater potential mapping. CATENA 182:104101. https://doi.org/10.1016/j.catena.2019.104101

Pourtaghi ZS, Pourghasemi HR (2014) GIS-based groundwater spring potential assessment and mapping in the Birjand Township, southern Khorasan Province Iran. Hydrogeol J 22:643–662. https://doi.org/10.1007/s10040-013-1089-6

Rahmati SHA, Zandieh M (2011) A new biogeography-based optimization (BBO) algorithm for the flexible job shop scheduling problem. Int J Adv Manuf Technol 58:1115–1129. https://doi.org/10.1007/s00170-011-3437-9

Rahmati O, Samani AN, Mahdavi M et al (2014) Groundwater potential mapping at Kurdistan region of Iran using analytic hierarchy process and GIS. Arab J Geosci 8:7059–7071. https://doi.org/10.1007/s12517-014-1668-4

Rahmati O, Pourghasemi HR, Melesse AM (2016) Application of GIS-based data driven random forest and maximum entropy models for groundwater potential mapping: a case study at Mehran region Iran. CATENA 137:360–372. https://doi.org/10.1016/j.catena.2015.10.010

Rajaeian S, Ketabchi H, Ebadi T (2024) Investigation on quantitative and qualitative changes of groundwater resources using MODFLOW and MT3DMS: a case study of Hashtgerd aquifer. Iran Environ Dev Sustain 26(2):4679–4704. https://doi.org/10.1007/s10668-022-02904-4

Rasool U, Yin X, Xu Z, Rasool MA, Senapathi V, Hussain M, Siddique J, Trabucco JC (2022) Mapping of groundwater productivity potential with machine learning algorithms: a case study in the provincial capital of Baluchistan. Pakistan Chemosphere 303:135265. https://doi.org/10.1016/j.chemosphere.2022.135265

Rasool U, Yin X, Xu Z et al (2022) Mapping of groundwater productivity potential with machine learning algorithms: a case study in the provincial capital of Baluchistan. Pakistan Chemosphere 303:135265. https://doi.org/10.1016/j.chemosphere.2022.135265

Razavi-Termeh SV, Sadeghi-Niaraki A, Choi S-M (2022) Spatio-temporal modelling of asthma-prone areas using a machine learning optimized with metaheuristic algorithms. Geocarto Int 37:9917–9942. https://doi.org/10.1080/10106049.2022.2028903

Razavi-Termeh SV, Seo M, Sadeghi-Niaraki A, Choi S-M (2023) Flash flood detection and susceptibility mapping in the Monsoon period by integration of optical and radar satellite imagery using an improvement of a sequential ensemble algorithm. Weather Climate Extremes 41:100595. https://doi.org/10.1016/j.wace.2023.100595

Sahin EK, Ipbuker C, Kavzoglu T (2016) Investigation of automatic feature weighting methods (Fisher, Chi-square and Relief-F) for landslide susceptibility mapping. Geocarto Int 32:956–977. https://doi.org/10.1080/10106049.2016.1170892

Santosa B, Safitri AL (2015) Biogeography-based optimization (BBO) algorithm for single machine total weighted tardiness problem (SMTWTP). Procedia Manuf 4:552–557. https://doi.org/10.1016/j.promfg.2015.11.075

Saqr AM, Nasr M, Fujii M, Yoshimura C, Ibrahim MG (2024) Delineating suitable zones for solar-based groundwater exploitation using multi-criteria analysis: A techno-economic assessment for meeting sustainable development goals (SDGs). Groundwater Sustain Dev 25:101087. https://doi.org/10.1016/j.gsd.2024.101087

Saqr AM, Nasr M, Fujii M, Yoshimura C, Ibrahim MG (2022a) Monitoring of agricultural expansion using hybrid classification method in southwestern fringes of Wadi El-Natrun, Egypt: an appraisal for sustainable development. In: Asia conference on environment and sustainable development. Springer, Singapore, pp 349–362

Saqr AM, Nasr M, Fujii M, Yoshimura C, Ibrahim MG (2022b) Optimal solution for increasing groundwater pumping by integrating MODFLOW-USG and particle swarm optimization Algorithm: a case study of Wadi El-Natrun, Egypt. In: International conference on environment science and engineering. Springer, Singapore, pp 59–73

Saremi S, Mirjalili S, Lewis A (2014) Biogeography-based optimisation with chaos. Neural Comput Appl 25:1077–1097. https://doi.org/10.1007/s00521-014-1597-x

Sarker IH (2021) Machine learning: algorithms, real-world applications and research directions. SN Comput Sci. https://doi.org/10.1007/s42979-021-00592-x

Saud MA (2010) Mapping potential areas for groundwater storage in Wadi Aurnah Basin, western Arabian Peninsula, using remote sensing and geographic information system techniques. Hydrogeol J 18:1481–1495. https://doi.org/10.1007/s10040-010-0598-9

Shrestha N (2020) Detecting multicollinearity in regression analysis. Am J Applied Math Stat 8:39–42. https://doi.org/10.12691/ajams-8-2-1

Simon D (2008) Biogeography-based optimization. IEEE Trans Evol Comput 12:702–713. https://doi.org/10.1109/tevc.2008.919004

Sobhani R, Emadi A, Fazloula R, Zamanzad-Ghavidel S (2024) Development of groundwater poverty index focusing on agricultural and industrial perspectives in West Azerbaijan Province. Iran Environ Dev Sustain 26(7):16555–16583. https://doi.org/10.1007/s10668-023-03299-6

Song Y, Li H, Xu P, Liu D (2022) A method of intrusion detection based on WOA-XGBoost algorithm. Discret Dyn Nat Soc 2022:1–9. https://doi.org/10.1155/2022/5245622

Song Y, Kalacska M, Gašparović M et al (2023) Advances in geocomputation and geospatial artificial intelligence (GeoAI) for mapping. Int J Appl Earth Obs Geoinf 120:103300. https://doi.org/10.1016/j.jag.2023.103300

Sreedevi PD, Subrahmanyam K, Ahmed S (2005) The significance of morphometric analysis for obtaining groundwater potential zones in a structurally controlled terrain. Environ Geol 47:412–420. https://doi.org/10.1007/s00254-004-1166-1

Su X, Lu S, Gao R et al (2017) Groundwater flow path determination during riverbank filtration affected by groundwater exploitation: a case study of Liao River, Northeast China. Hydrol Sci J 62:2331–2347. https://doi.org/10.1080/02626667.2017.1383609

Tegegne AM, Lohani TK, Eshete AA (2024) Groundwater potential delineation using geodetector based convolutional neural network in the Gunabay watershed of Ethiopia. Environ Res 242:117790. https://doi.org/10.1016/j.envres.2023.117790

Teimouri H, Ansari M, Teimouri H et al (2018) Assessing potential of physical development with an emphasis on geomorphological indicators using AHP-FUZZY (Case Study: Estahban City). Asian J Water Environ Pollut 15:115–126. https://doi.org/10.3233/ajw-180024

Tran DA, Tsujimura M, Ha NT et al (2021) Evaluating the predictive power of different machine learning algorithms for groundwater salinity prediction of multi-layer coastal aquifers in the Mekong Delta. Vietnam Ecol Indicators 127:107790. https://doi.org/10.1016/j.ecolind.2021.107790

Upwanshi M, Damry K, Pathak D, Tikle S, Das S (2023) Delineation of potential groundwater recharge zones using remote sensing, GIS, and AHP approaches. Urban Climate 48:101415. https://doi.org/10.1016/j.uclim.2023.101415

Vafadar S, Rahimzadegan M, Asadi R (2023) Evaluating the performance of machine learning methods and Geographic Information System (GIS) in identifying groundwater potential zones in Tehran-Karaj plain Iran. J Hydrol 624:129952. https://doi.org/10.1016/j.jhydrol.2023.129952

Van Phong T, Pham BT, Trinh PT et al (2021) Groundwater potential mapping using GIS-based hybrid artificial intelligence methods. Ground Water 59:745–760. https://doi.org/10.1111/gwat.13094

Wu S, Yuan Q, Yan Z, Xu Q (2021) Analyzing accident injury severity via an extreme gradient boosting (XGBoost) model. J Adv Transp 2021:1–11. https://doi.org/10.1155/2021/3771640

Xiong H, Guo X, Wang Y, Xiong R, Gui X, Hu X, Li Y, Qiu Y, Tan J, Ma C (2023) Spatial prediction of groundwater potential by various novel boosting-based ensemble learning models in mountainous areas. Geocarto Int. https://doi.org/10.1080/10106049.2023.2274870

Yadav B, Malav LC, Jangir A et al (2023) Application of analytical hierarchical process, multi-influencing factor, and geospatial techniques for groundwater potential zonation in a semi-arid region of western India. J Contam Hydrol 253:104122. https://doi.org/10.1016/j.jconhyd.2022.104122

Yang L, Shami A (2020) On hyperparameter optimization of machine learning algorithms: theory and practice. Neurocomputing 415:295–316. https://doi.org/10.1016/j.neucom.2020.07.061

Yariyan P, Avand M, Omidvar E et al (2022) Optimization of statistical and machine learning hybrid models for groundwater potential mapping. Geocarto Int 37:3877–3911. https://doi.org/10.1080/10106049.2020.1870164

Zamani MG, Moridi A, Yazdi J (2022) Groundwater management in arid and semi-arid regions. Arabian J Geosci. https://doi.org/10.1007/s12517-022-09546-w

Zhang X, Yan C, Gao C et al (2020) Predicting missing values in medical data Via XGBoost regression. J Healthcare Inf Res 4:383–394. https://doi.org/10.1007/s41666-020-00077-1

Zheng Y, Lu X, Zhang M, Chen S (2019) Biogeography-based optimization: algorithms and applications

Zhong S, Zhang K, Bagheri M et al (2021) Machine learning: new ideas and tools in environmental science and engineering. Environ Sci Technol. https://doi.org/10.1021/acs.est.1c01339

Zhu X, Liu H, He W, Wu L, Liu F (2023) Regolith water storage patterns determine vegetation productivity in global karst regions. Geoderma 430:116292. https://doi.org/10.1016/j.geoderma.2022.116292

Publisher's Note Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations" (in PDF at the end of the article below the references; in XML as a back matter article note).